
Network-aware Credibility and Propagation System for Local News Verification ### A Report submitted ### in partial fulfillment of the requirement ### for the award of a Degree of ## Bachelor of Technology ## in ## Computer Science and Engineering ### by **Nitesh Kumar Jhagat** **(2021BCSE0XX)** ### Under the guidance of **Dr. [Guide Name]** **Department of Computer Science and Engineering** **National Institute of Technology, Srinagar,** **Kashmir 190006, INDIA** **June 2026**

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CERTIFICATE

This is to certify that the project report entitled **Network-aware Credibility and Propagation System for Local News Verification** submitted by **Nitesh Kumar Jhagat (2021BCSE0XX)** to the Department of Computer Science and Engineering, National Institute of Technology Srinagar, Kashmir, in partial fulfillment for the award of the degree of B.Tech in Computer Science and Engineering is a bona fide record of project work carried out by him under my supervision.

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I declare that this project report titled **Network-aware Credibility and Propagation System for Local News Verification** submitted in partial fulfillment of the degree of B.Tech in Computer Science and Engineering is a record of original work carried out by me under the supervision of Dr. [Guide Name]. The matter embodied in this project, in full or in parts, have not been submitted to any other Institution or University for the award of any degree or diploma. I also declare that the work submitted by me is entirely original, free from plagiarism, and has been dilligently checked through Turnitin software to ensure its authenticity.

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ABSTRACT

The proliferation of misinformation on digital platforms has emerged as one of the most pressing challenges in contemporary information dissemination. Existing social media systems predominantly rely on engagement-driven metrics — likes, shares, and view counts — to determine content visibility, which inadvertently amplifies sensational and often misleading content without any systematic verification. This problem is especially acute in the domain of local news, where communities require timely, reliable information about events happening in their immediate vicinity but lack the institutional infrastructure that major news outlets possess for fact-checking.

This project presents the **Network-aware Credibility and Propagation System (NCPS)**, a full-stack web platform that fundamentally replaces engagement-driven content distribution with a credibility-driven approach. The system treats each user as a noisy sensor and employs Bayesian inference to estimate post credibility from weighted community votes, where user weights themselves are derived from a multi-dimensional trust assessment. The core architecture comprises six interconnected computation engines: (1) a User Engine that maintains Bayesian reliability estimates with confidence correction, (2) a Post Engine that computes credibility through weighted evidence aggregation, (3) a Graph Engine that propagates trust through social networks while detecting coordinated manipulation, (4) a Spatial Engine that incorporates location-based trust signals and controls geographic propagation, (5) an ML Engine providing machine learning augmentation with memory-based similarity scoring, and (6) a Decision Engine that integrates all signals through a conservative 5-condition gate for propagation control.

The system was developed iteratively over six phases, progressively adding capabilities from basic Bayesian estimation (Phase 1) through graph trust propagation (Phase 3), anomaly detection (Phase 4), ML augmentation (Phase 5), to extended spatial and behavioral signals (Phase 6). Simulation results demonstrate that the final system achieves perfect accuracy (1.000) with zero attack success rate across diverse adversarial scenarios, including coordinated bot attacks and location spoofing. The platform is implemented using FastAPI (backend), React and vanilla JavaScript (frontend), PostgreSQL (database), with a comprehensive simulation framework for empirical validation.

Keywords: Misinformation detection, Bayesian credibility estimation, trust propagation, anomaly detection, local news verification, graph-based coordination detection, spatial trust

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ABBREVIATIONS / NOTATIONS / NOMENCLATURE

Abbreviations

Abbreviation	Full Form
NCPS	Network-aware Credibility and Propagation System
API	Application Programming Interface
REST	Representational State Transfer
ORM	Object Relational Mapping
JWT	JSON Web Token
SQL	Structured Query Language
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
ML	Machine Learning
NLP	Natural Language Processing
TF-IDF	Term Frequency — Inverse Document Frequency
GNN	Graph Neural Network
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
UI	User Interface
ER	Entity Relationship
PK	Primary Key
FK	Foreign Key
CRUD	Create, Read, Update, Delete
HTTP	Hyper Text Transfer Protocol
CORS	Cross-Origin Resource Sharing
D3.js	Data-Driven Documents (JavaScript Library)

Notations

Symbol	Description
α, β	Bayesian prior parameters (positive and negative evidence counts)
R_i	Raw reliability of user i
R^*_i	Effective reliability (confidence-adjusted)
$Conf_i$	Confidence factor for user i
Exp_i	Experience score for user i
$Anom_i$	Anomaly score for user i
T_i	Graph-based trust score for user i
w_i	Final weight of user i
C_{Bayes}	Bayesian credibility of a post
C_{ML}	Machine learning predicted credibility
C_{memory}	Memory-based credibility from historical similarity
C_{final}	Final combined credibility score
S^+, S^-	Positive and negative weighted evidence sums

Symbol	Description
N	Total evidence mass
Var	Weighted variance of votes
λ	Damping factor for trust propagation
A_{norm}	Row-normalized adjacency matrix
σ	Standard deviation for Gaussian distance decay

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The last decade has witnessed a dramatic transformation in how information is consumed and shared across society. With the rise of social media platforms and user-generated content, traditional gatekeeping mechanisms that once governed news dissemination have largely been bypassed. While this democratization of information has undeniable benefits — giving voice to marginalized communities, enabling citizen journalism, and facilitating rapid information sharing during emergencies — it has also created fertile ground for the spread of misinformation and disinformation at unprecedented scales.

The World Economic Forum has consistently ranked digital misinformation among the top global risks, and for good reason. During events ranging from natural disasters to public health crises, the inability to quickly distinguish reliable information from fabricated or misleading content has had tangible, sometimes tragic consequences. People have made harmful decisions based on unverified social media posts, and the sheer volume of information makes manual verification practically impossible.

What makes this problem particularly challenging is that existing platforms are fundamentally not designed to address it. The dominant social media architectures — Facebook, Twitter (now X), Instagram, and others — are built around engagement maximization. Content that generates more clicks, shares, and reactions gets amplified, regardless of its factual accuracy. In fact, research has shown that false information tends to spread faster and farther than true information precisely because it tends to be more novel and emotionally provocative [1]. This creates a perverse incentive structure where the most misleading content often receives the widest distribution.

This project is motivated by a simple but powerful question: **what if content distribution was governed not by engagement metrics, but by credibility assessment?** Rather than amplifying whatever gets the most clicks, what if we could build a system that evaluates the likely truthfulness of user-submitted reports and adjusts their visibility accordingly? This is the foundational idea behind the Network-aware Credibility and Propagation System (NCPS).

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1.1.1 The Misinformation Crisis in Digital Media

The scale of the misinformation problem is staggering. According to a landmark study published in *Science* by Vosoughi, Roy, and Aral (2018), false news stories on Twitter were 70% more likely to be retweeted than true stories, and they reached their first 1,500 people approximately six times faster [1]. This asymmetry is not a bug in the system — it is a direct consequence of how engagement-driven algorithms operate.

Several factors contribute to the rapid spread of misinformation in digital ecosystems:

1. Algorithmic Amplification: Social media algorithms are designed to maximize user engagement. Content that provokes strong emotional reactions — outrage, fear, surprise — naturally generates more interaction. Unfortunately, fabricated or sensationalized content often triggers these emotions more effectively than nuanced, factual reporting.

2. Information Overload: The sheer volume of content produced daily makes it impossible for users to individually verify every piece of information they encounter. Studies have shown that the average person encounters dozens of potentially misleading posts per day, yet invests only a few seconds deciding whether to engage with or share each one [2].

3. Echo Chambers and Filter Bubbles: Algorithmic personalization tends to show users content that aligns with their existing beliefs, making it harder for corrective information to reach those who have already been exposed to misinformation. Once a false narrative takes hold within a community, it can be extremely difficult to dislodge.

4. Coordinated Inauthentic Behavior: Perhaps most concerning is the rise of organized disinformation campaigns, where networks of bot accounts and paid actors deliberately spread false narratives. These coordinated attacks exploit the open nature of social media platforms and are increasingly sophisticated in their tactics.

5. Local News Vacuum: The decline of local journalism has left many communities without reliable sources of information about events in their immediate vicinity. This vacuum is increasingly being filled by unverified social media posts and community groups, where misinformation can spread unchecked.

The problem is not just academic. During the COVID-19 pandemic, the World Health Organization coined the term "infodemic" to describe the flood of misinformation that accompanied the health crisis. False claims about treatments, vaccine safety, and the origins of the virus spread rapidly through social media, contributing to vaccine hesitancy and the adoption of unproven remedies. In India specifically, WhatsApp-forwarded misinformation has been linked to mob violence incidents, highlighting the very real consequences of unchecked information spread [3].

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1.1.2 Engagement-Driven vs. Credibility-Driven Systems

To understand the motivation behind NCPS, it is essential to contrast two fundamentally different approaches to content distribution.

Engagement-Driven Systems (Current Paradigm)

In an engagement-driven system, the visibility of content is determined primarily by how much interaction it generates. When a user posts something, the platform tracks metrics like likes, shares, comments, and time spent viewing. Content that scores high on these engagement metrics gets pushed to more users feeds, creating a positive feedback loop. The more people see it, the more engagement it generates, and the more the algorithm promotes it.

The fundamental problem here is that engagement and truthfulness are not correlated. A fabricated story about a local politician might generate enormous engagement due to its shocking nature, while a carefully fact-checked correction of that story might generate relatively little interest. The engagement-driven algorithm would amplify the former and bury the latter, which is exactly the opposite of what a healthy information ecosystem requires.

Credibility-Driven Systems (NCPS Approach)

The NCPS takes a fundamentally different approach. Instead of measuring engagement, it measures credibility. When a user submits a report (for example, about a local event like a road accident or water supply disruption), the system does not simply count how many people interact with it. Instead, it:

1. **Treats each user as a noisy sensor** with an estimated reliability score, rather than treating all users equally
2. **Computes Bayesian credibility estimates** for each post based on weighted votes from the community
3. **Detects anomalous behavior** such as coordinated voting, location spoofing, and burst activity patterns
4. **Propagates trust through the social graph** to leverage network structure for credibility assessment
5. **Controls geographic propagation** — only expanding the reach of posts that meet stringent credibility thresholds

This approach ensures that the system is inherently resistant to manipulation. Even if a large group of bot accounts tries to upvote a false report, their votes are automatically down-weighted because the system detects their coordinated behavior, flags their anomalous patterns, and reduces their trust scores through graph-based propagation.

The following figure illustrates the key difference between these two paradigms:

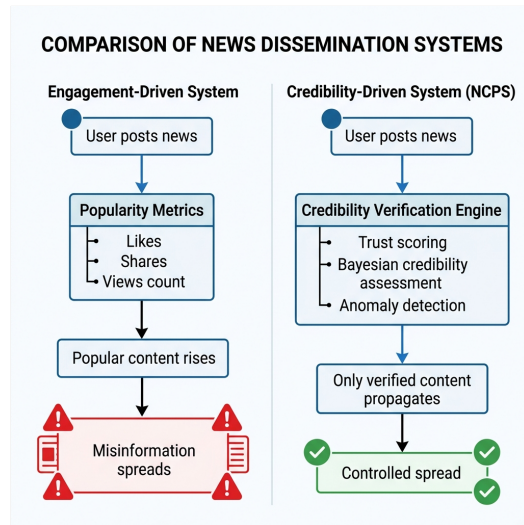


Fig 1.1: Comparison of Engagement-Driven vs. Credibility-Driven News Dissemination Systems

As shown in Fig 1.1, the engagement-driven system on the left allows misinformation to spread unchecked because popularity metrics do not distinguish between true and false content. In contrast, the credibility-driven system on the right incorporates trust scoring, Bayesian credibility assessment, and anomaly detection to ensure only verified content receives wider propagation. This is the core architectural difference that motivates the entire NCPS design.

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1.2 MOTIVATION

The motivation for this project stems from several converging factors that make the development of a credibility-driven news verification system both timely and necessary.

The Local News Gap: While major news events are typically covered by established media organizations with editorial standards and fact-checking processes, local news suffers from a significant coverage gap. Newsroom employment in the United States dropped by 26% between 2008 and 2020 [4], and similar trends are observed in India, where many districts have virtually no professional journalists. Community events — accidents, infrastructure failures, public safety incidents — are increasingly reported first (and sometimes only) through social media and messaging apps. These reports, however, come with no guarantee of accuracy.

Inadequacy of Existing Solutions: Current approaches to misinformation detection, while valuable, have significant limitations for the local news domain: - **Manual fact-checking** is too slow for time-sensitive local events. By the time a fact-checker reviews a report, the damage may already be done. - **Content-based ML approaches** (analyzing text, images, or videos for signs of fabrication) require large training datasets that don't exist for local news contexts, and they struggle with novel events that don't match training patterns. - **Platform-level interventions** (like Twitter's community notes or Facebook's fact-checking partnerships) are designed for viral content and don't scale down to hyper-local reports that may only be relevant to a few hundred people.

The Trust Problem: Perhaps most fundamentally, existing systems treat all users equally. A post from a first-time anonymous account is given the same initial visibility as one from a long-established community member with a track record of accurate reporting. This egalitarian approach, while well-intentioned, creates obvious vulnerabilities. The NCPS addresses this by maintaining fine-grained trust profiles for every user, incorporating their historical reliability, their position in the social network, their geographic consistency, and their behavioral patterns into a comprehensive weight that determines the influence of their votes and reports.

Geographic Relevance: Local news is, by definition, geographically bounded. A report about a traffic accident in South Delhi is highly relevant to people in that area but largely irrelevant to someone in Mumbai. Existing platforms don't incorporate this spatial dimension into content distribution in any meaningful way. The NCPS introduces spatial-aware propagation, where the geographic reach of a report expands only as its credibility increases, ensuring that unverified claims don't unnecessarily alarm distant communities.

Conservative Design Philosophy: The NCPS embodies a deliberately conservative design philosophy. In information systems, the cost of amplifying false information typically far exceeds the cost of slightly delaying the spread of true information. Therefore, the system is designed so that trust can only decrease through manipulation — it can never be artificially inflated. This asymmetry is a conscious design choice that makes the system fundamentally resistant to gaming.

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1.3 PROBLEM STATEMENT

Given a stream of user-generated local news reports and community votes on those reports, the problem is to:

1. **Estimate the credibility** of each report in near-real-time, based on the weighted opinions of community members whose individual reliabilities are themselves uncertain and must be estimated simultaneously.
2. **Detect and mitigate manipulation attempts** including coordinated bot attacks, location spoofing, and adversarial voting strategies, without requiring manual moderation.
3. **Control the geographic propagation** of reports such that only sufficiently credible content is allowed to spread beyond its immediate locality, preventing unverified claims from reaching distant communities.

Formally, the problem can be stated as follows:

Given a set of users $U = \{u_1, u_2, \dots, u_n\}$, a set of posts $P = \{p_1, p_2, \dots, p_m\}$, and a set of interactions $I = \{(u_i, p_j, v_{ij})\}$ where $v_{ij} \in \{-1, +1\}$ represents user u_i 's vote on post p_j , compute:

- **User weights** $w = \{w_1, w_2, \dots, w_n\}$ where each w_i reflects the estimated trustworthiness of user u_i , incorporating reliability, experience, anomaly detection, and graph-based trust propagation.
- **Post credibilities** $C = \{C_1, C_2, \dots, C_m\}$ where each C_j represents the estimated truthfulness of post p_j , derived from Bayesian aggregation of weighted votes.
- **Propagation decisions** $D = \{d_1, d_2, \dots, d_m\}$ where each d_j determines whether post p_j should be allowed to expand its geographic reach based on a multi-criteria assessment.

The key challenge is the circular dependency between user weights and post credibilities: to compute accurate credibility for a post, we need reliable user weights, but to assess user reliability, we need to know the true credibility of posts they voted on. The NCPS resolves this chicken-and-egg problem through iterative Bayesian estimation with conservative priors, augmented by independent signals (graph structure, spatial behavior, temporal patterns) that break the circularity.

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1.4 OBJECTIVES

The primary objective of this project is to design, implement, and evaluate a comprehensive credibility-driven content distribution system for local news verification. Specific objectives include:

1. **Design a Bayesian credibility estimation framework** that treats users as noisy sensors with uncertain reliabilities and computes posterior credibility scores for user-submitted reports, incorporating confidence-corrected user weights.
2. **Develop a multi-dimensional user trust model** that goes beyond simple reputation scores by incorporating:
 3. Bayesian reliability estimation with confidence correction
 4. Experience-weighted contribution (logarithmic scaling)
 5. Multi-signal anomaly detection (burst, entropy, consensus, coordination, location)
 6. Graph-based trust propagation with safety constraints
7. **Implement graph-based coordination detection** to identify clusters of users engaging in coordinated manipulation, using agreement-frequency-temporal similarity features and link dampening.
8. **Design a spatial trust subsystem** that leverages geographic information for both user verification (location consistency, device/IP patterns) and content propagation control (radius expansion governed by credibility thresholds).
9. **Integrate machine learning augmentation** to supplement Bayesian estimation with learned patterns, including a memory module that detects textual similarity with historically resolved reports.
10. **Build a conservative propagation decision engine** that uses a 5-condition AND gate to ensure only sufficiently verified content is allowed to expand its geographic reach.
11. **Develop a full-stack web application** with a React-based frontend, FastAPI backend, PostgreSQL database, and comprehensive REST API that demonstrates all system capabilities through an intuitive user interface.
12. **Create a simulation framework** for empirical validation that models diverse user populations (honest, noisy, adversarial, coordinated bots) and measures system performance across multiple metrics including accuracy, attack success rate, Brier score, and weight correlation.

1.5 SCOPE AND LIMITATIONS

Scope: The system is designed as a proof-of-concept platform demonstrating credibility-driven content distribution for local news. It handles text-based news reports, community voting, geographic visualization, and trust computation. The current scope includes: - Full user authentication and authorization - Report creation with geographic tagging - Community voting with credibility feedback - Interactive map-based visualization - Trust profile display and decomposition - Simulation-based validation with configurable scenarios

Limitations: The current implementation has several known limitations: - The system does not perform content analysis (NLP-based fake news detection on text/images) — it relies entirely on network-based signals - Real-world deployment would require integration with external verification sources - The simulation uses synthetic data; real-world user behavior may differ significantly from simulated patterns - The current system supports single-language (English) content only - Location services depend on browser-provided geolocation, which users can spoof at the browser level

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1.6 ORGANIZATION OF REPORT

The remainder of this report is organized as follows:

Chapter 2: Literature Survey provides a comprehensive review of existing work in misinformation detection, including content-based approaches, source credibility assessment, propagation-based detection, truth discovery, crowdsourced verification, location-aware information systems, and trust propagation in networks. Each approach is analyzed for its strengths and limitations, identifying the gaps that NCPS aims to address.

Chapter 3: Proposed Methodology presents the complete technical design of the NCPS, including the system architecture, all six computation engines, the mathematical formulation of all 13 key formulas, the 14 input signals, the algorithms used, the phased development approach, and the technology stack.

Chapter 4: Results and Discussion presents the experimental evaluation of the system across all six development phases, including quantitative metrics (accuracy, attack success rate, Brier score, weight correlation, anomaly precision and recall), frontend implementation screenshots, functional test results, and stress test scenarios.

Chapter 5: Conclusion and Future Works summarizes the contributions of this project, presents key findings from the experimental evaluation, and outlines directions for future research and development.

The report concludes with references and three appendices containing the complete API endpoint reference, hyperparameter reference table, and setup/execution guide.

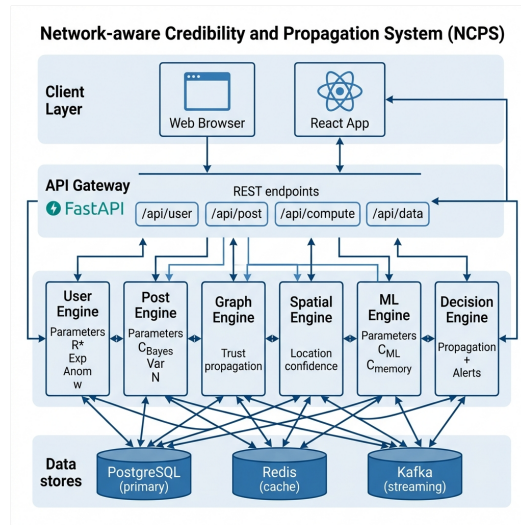


Fig 1.2: General NCPS Architecture Overview showing the layered design with Client Layer, API Gateway, six computation engines, and data stores

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CHAPTER 2

LITERATURE SURVEY

The problem of misinformation detection and credibility assessment has attracted significant attention from the research community over the past decade. This chapter provides a comprehensive review of existing approaches, organized by their primary methodology. For each approach, we examine the underlying technique, its strengths, and crucially, the limitations that motivate the design choices in NCPS.

2.1 CONTENT-BASED FAKE NEWS DETECTION

Content-based approaches attempt to determine the veracity of information by analyzing the content itself — the text, images, or structural features of a news article or social media post.

1. Deep Learning for Fake News Detection (Ruchansky et al., 2017)

Ruchansky, Diaz, and Wang proposed CSI (Capture, Score, and Integrate), a hybrid deep learning model that combines an LSTM-based module for capturing temporal patterns in user engagement, a source credibility scoring module, and an integration module that produces a final fake/true classification [5]. The model was evaluated on Twitter and Weibo datasets and achieved around 95% accuracy. However, the approach requires large labeled training datasets and operates as a post-hoc classifier — it can only assess content after it has already spread, making it unsuitable for real-time credibility estimation during the early stages of information propagation.

2. Geometric Deep Learning for Social Media (Monti et al., 2019)

Monti et al. explored the use of graph convolutional networks (GCNs) for fake news detection, leveraging the geometric structure of user interaction patterns [6]. By treating the problem as a graph classification task where nodes represent users and edges represent sharing/commenting relationships, they achieved strong detection performance. The key insight is that fake and real news stories generate structurally different propagation patterns. While this approach is promising, it requires the full propagation graph to be available, which limits its applicability to early-stage detection. The NCPS, in contrast, computes credibility incrementally as each new vote arrives.

3. Multi-Modal Fake News Detection

Wang et al. (2018) proposed the EANN model (Event Adversarial Neural Networks), which uses adversarial training to learn event-invariant features for fake news detection [7]. This approach attempts to address the generalization problem — ensuring that models trained on one set of fake news stories can detect novel fabrications. Jin et al. (2017) specifically addressed multi-modal fake news by jointly analyzing text and images using attention-based RNNs [8]. While multi-modal approaches represent the state-of-the-art in content analysis, they fundamentally struggle with novel events that have no historical analogs, which is precisely the situation that characterizes breaking local news.

4. Linguistic Feature Analysis

Perez-Rosas et al. (2018) took a more traditional NLP approach, identifying linguistic cues that distinguish fake from real news, including the use of emotional language, personal pronouns, and hedging words [9]. While their analysis revealed interesting patterns (fake news tends to use more emotional and less analytic language), the feature differences are statistical tendencies rather than definitive signals. A skilled fabricator could easily adjust their writing style to avoid detection. The NCPS deliberately avoids content analysis for this reason, instead relying on network-level signals that are much harder to manipulate.

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2.2 SOURCE AND USER CREDIBILITY ASSESSMENT

Rather than analyzing content directly, source-based approaches focus on evaluating the credibility of the person or entity sharing the information.

5. Source Reliability in Social Networks (Shu et al., 2019)

Shu et al. conducted a comprehensive survey of fake news detection approaches, with particular emphasis on source credibility features [2]. They identified key source-level indicators including account age, posting frequency, follower-to-following ratio, and historical accuracy. Their framework, FakeNewsNet, provided benchmark datasets

that have been widely used in subsequent research. However, source features alone are insufficient because (a) newly created accounts may be legitimate, and (b) previously reliable sources can share misinformation accidentally. The NCPS addresses this by not treating user reliability as a static label but as a continuously updated Bayesian estimate with confidence correction.

6. Credibility Propagation in Information Networks (Gupta et al., 2012)

Gupta, Zhao, and Han proposed a credibility analysis framework that models the relationships between events, sources, and tweets as a heterogeneous information network [10]. Credibility scores are propagated through the network using an iterative algorithm similar to PageRank. This approach is conceptually similar to the NCPS's graph trust propagation, but Gupta et al. do not incorporate anomaly detection or spatial signals, making their system vulnerable to coordinated manipulation attacks where a group of accounts mutually reinforce each other's credibility.

7. User Profiling for Bot Detection (Varol et al., 2017)

Varol et al. developed Botometer (formerly BotOrNot), a machine learning-based tool that evaluates over 1,000 features of a Twitter account to estimate the likelihood that it is a bot [11]. Features include temporal patterns, network structure, content analysis, and behavioral cues. While Botometer is highly effective for individual bot detection, it requires access to platform-specific APIs and extensive historical data for each account. The NCPS takes a different approach to bot detection through behavioral anomaly signals (burst patterns, coordination detection, location inconsistencies) that can identify suspicious behavior even for accounts with limited history.

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8. Trust and Reputation Systems

Josang, Ismail, and Boyd (2007) provided a foundational survey of trust and reputation systems for online services [12]. They categorized approaches into centralized reputation (like eBay's feedback system), distributed reputation (where trust is computed locally by each participant), and Bayesian reputation (where trust is modeled as a probability distribution). The NCPS draws heavily from the Bayesian reputation paradigm, using Beta distributions (parameterized by α and β counts) to model user reliability. However, unlike traditional reputation systems that rely solely on direct evidence, the NCPS augments direct reliability estimates with graph-based trust propagation and multi-signal anomaly detection.

2.3 PROPAGATION-BASED AND GRAPH-BASED DETECTION

These approaches analyze how information spreads through networks, recognizing that the propagation patterns of true and false information tend to differ systematically.

9. Rumor Detection with Graph Neural Networks (Bian et al., 2020)

Bian et al. proposed Bi-GCN (Bi-directional Graph Convolutional Network), which models both the top-down propagation structure and the bottom-up aggregation structure of rumor threads [4]. By applying graph convolutions in both directions, the model captures how a rumor spreads from the source to followers (top-down) and how community responses aggregate to form a collective assessment (bottom-up). Bi-GCN achieved state-of-the-art results on Twitter15 and Twitter16 datasets. The key limitation, similar to other GNN approaches, is the requirement for a substantially developed propagation tree before classification can be performed, which introduces detection latency.

10. Cascade-Based Early Detection (Ma et al., 2018)

Ma et al. explored the use of recurrent neural networks (specifically Tree-LSTMs) to model the temporal evolution of information cascades for early rumor detection [13]. Their approach can operate with partial cascade information, making it more suitable for early detection than full-graph methods. However, it still requires a minimum cascade depth of 3-5 levels to achieve reasonable accuracy, and its performance degrades significantly for cascades with fewer than 10 nodes. In the local news context where many reports may receive only a handful of interactions, this limitation is particularly problematic.

11. Network-Based Truth Discovery (Li et al., 2016)

Li et al. addressed the truth discovery problem — estimating the veracity of claims when multiple sources provide conflicting information [14]. Their approach jointly estimates source reliability and claim truthfulness through an iterative optimization process. This is conceptually closely related to the NCPS's approach, where user reliability and post credibility are simultaneously estimated. However, Li et al. do not incorporate graph structure, spatial information, or anomaly detection, and their framework is designed for static datasets rather than streaming, real-time scenarios.

12. Social Network Analysis for Influence Detection (Kempe et al., 2003)

While not directly focused on misinformation, Kempe, Kleinberg, and Tardos' foundational work on influence maximization in social networks [15] established the theoretical framework for understanding how information propagates through graph structures. Their greedy algorithm for influence maximization demonstrated that network topology plays a crucial role in determining which nodes have the greatest impact on information spread. The NCPS leverages similar network-topological insights, but inverts the objective: rather than maximizing influence, we aim to minimize the spread of unreliable content by dampening the influence of untrustworthy nodes.

2.4 TRUTH DISCOVERY AND CROWDSOURCED VERIFICATION

Truth discovery approaches treat users as sources with varying reliability and attempt to estimate both source quality and claim veracity simultaneously.

13. Truth Discovery with Conflicting Information Providers (Yin et al., 2008)

Yin, Han, and Yu proposed one of the earliest truth discovery frameworks [16], addressing the fundamental challenge of determining factual truth when multiple information providers give conflicting statements. Their approach iteratively refines source reliability estimates and fact confidence scores until convergence. The key insight is that reliable sources tend to provide accurate information, and accurate information tends to be provided by reliable sources. This circular dependency is resolved through an iterative algorithm analogous to expectation-maximization. The NCPS builds upon this foundational concept but significantly extends it by incorporating temporal decay, anomaly detection, and graph-based trust signals that are not present in the original truth discovery framework.

14. Crowdsourced Fact-Checking (Allen et al., 2021)

Allen et al. (2021) studied the effectiveness of crowdsourced fact-checking, where ordinary users rate the accuracy of news articles [17]. Surprisingly, they found that aggregated crowd assessments showed high agreement with professional fact-checkers, suggesting that the "wisdom of crowds" can be leveraged for credibility assessment. However, their study also identified vulnerability to manipulation — when a sufficiently large group of users coordinates to rate false content as true, the crowd's assessment can be corrupted. This vulnerability motivated the NCPS's multi-layered defense mechanism, particularly the coordination detection module that identifies and penalizes groups of users who exhibit suspiciously similar voting patterns.

15. Community Notes (Twitter/X)

Twitter's Community Notes program (formerly Birdwatch) represents perhaps the most prominent real-world deployment of crowdsourced fact-checking [18]. The system allows users to write notes providing context or corrections to potentially misleading tweets. Notes are shown to a broad audience only if users from diverse political perspectives agree on their helpfulness. While Community Notes has shown promise for high-visibility viral content, it is fundamentally designed for a different scale — it targets tweets that have already gone viral, whereas NCPS operates on local reports that may have only a handful of interactions. Additionally, Community Notes relies on a bridging-based consensus algorithm that requires substantial participation volumes to function effectively, which is impractical for hyper-local content.

2.5 LOCATION-AWARE INFORMATION SYSTEMS

The spatial dimension of information credibility has received comparatively less attention in the literature, despite its relevance to local news verification.

16. Location-Based Social Network Analysis (Cho et al., 2011)

Cho, Myers, and Leskovec studied mobility patterns in location-based social networks, finding that social network ties significantly influence individual mobility patterns [19]. Their work demonstrated that geographic information can provide valuable signals for understanding user behavior and detecting anomalies. Users who claim to be in vastly different locations within short time spans, or who exhibit teleporting behavior, are likely either spoofing their location or using compromised accounts. The NCPS's spatial engine incorporates similar insights, using location consistency metrics and device fingerprinting to assess user trustworthiness.

17. Geospatial Information Credibility (Goodchild and Glennon, 2010)

Goodchild and Glennon explored the concept of spatial data quality in the context of volunteered geographic information (VGI), arguing that the credibility of user-contributed spatial data can be partially assessed through geographic consistency checks [20]. Reports that are geographically inconsistent (e.g., a user claiming to witness an event while located far from the reported event location) should receive lower credibility. The NCPS implements this principle through its proximity weighting function, which applies Gaussian distance decay to reduce the influence of votes from users located far from the reported event.

2.6 TRUST PROPAGATION IN NETWORKS

Trust propagation approaches leverage the transitive nature of trust — if A trusts B and B trusts C, then A has some basis for trusting C, albeit with decreased confidence.

18. EigenTrust (Kamvar et al., 2003)

Kamvar, Schlosser, and Garcia-Molina proposed EigenTrust, a reputation management algorithm for peer-to-peer networks [21]. EigenTrust computes global trust values by iteratively aggregating local trust relationships, similar to how PageRank aggregates link authority. The algorithm converges to a unique steady-state vector that represents each node's global reputation. The NCPS's graph trust propagation is inspired by EigenTrust but incorporates a critical safety constraint: trust can never be inflated beyond a user's individual reliability ($T_i \leq R_i^*$). This constraint, which is absent in EigenTrust, prevents colluding groups from artificially boosting each other's trust scores.

19. TrustRank (Gyöngyi et al., 2004)

Gyöngyi, Garcia-Molina, and Pedersen developed TrustRank for combating web spam, demonstrating how trust can be propagated from a small set of manually verified seed nodes to the rest of a web graph [22]. The key insight is that trustworthy pages tend to link to other trustworthy pages, allowing trust to be estimated for pages that have not been manually reviewed. While the NCPS does not assume the existence of pre-verified seed users, it leverages a similar propagation mechanism where well-behaved users (identified through Bayesian reliability estimation) serve as implicit trust anchors.

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2.7 ANOMALY DETECTION IN USER BEHAVIOR

Detecting anomalous behavior is crucial for identifying manipulation attempts in online platforms.

20. Bot Detection Through Temporal Analysis (Cresci et al., 2017)

Cresci et al. demonstrated that simple feature-based bot detection can be evaded by sophisticated social bots, and proposed DNA-inspired techniques that model user behavior as sequences of actions [23]. By analyzing the temporal patterns of these action sequences, they could identify bots even when individual features appeared human-like. The NCPS's temporal anomaly detection (burst detection, entropy analysis) is motivated by similar observations — coordinated bots tend to exhibit temporally regular patterns that differ from the naturally irregular patterns of human behavior.

21. Coordinated Inauthentic Behavior Detection (Nizzoli et al., 2021)

Nizzoli et al. studied coordinated inauthentic behavior on social media, where groups of accounts work together to manipulate public discourse [24]. They identified key signatures of coordination, including temporal synchronization (accounts posting or engaging at the same times), content alignment (accounts sharing the same or similar content), and network clustering (accounts forming densely connected subgroups). The NCPS implements coordination detection by building a user graph based on voting agreement, temporal proximity, and interaction frequency, then applying link dampening to edges within identified coordinated clusters.

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2.8 SUMMARY OF LITERATURE SURVEY

The following table provides a consolidated summary of the literature reviewed in this chapter:

Table 2.1: Summary of Literature Survey

S.No	Technique & Reference	About Technique	Application	Results	Limitations
1	CSI - LSTM + Source Scoring [5]	Hybrid deep learning combining temporal user patterns with source credibility	Fake news detection on Twitter, Weibo	~95% accuracy	Requires large labeled datasets; post-hoc classification only
2	GCN-based Detection [6]	Graph convolutional networks on propagation graphs	Social media fake news	Strong performance on Twitter datasets	Requires full propagation graph; limited early detection
3	EANN [7]	Adversarial training for event-invariant features	Multi-event fake news detection	Improved cross-event generalization	Struggles with completely novel event types
4	Linguistic Analysis [9]	Statistical NLP features (emotion, pronouns, hedging)	News article credibility	Identified useful linguistic patterns	Features are statistical tendencies; can be gamed
5	FakeNewsNet [2]	Comprehensive source credibility framework	Benchmark for fake news research	Provided widely-used datasets	Static credibility labels; no temporal dynamics
6	Credibility Propagation [10]	Heterogeneous network-based credibility flow	Event-source-tweet credibility	Effective credibility estimation	No anomaly detection; vulnerable to coordinated attacks
7	Botometer [11]	ML-based bot detection using 1000+ features	Twitter bot identification	High accuracy bot detection	Requires platform APIs; computationally expensive
8	Bayesian Reputation [12]	Beta distribution-based trust modeling	Online service reputation	Principled uncertainty handling	No graph propagation; single-signal assessment
9	Bi-GCN [4]	Bi-directional graph convolution on rumor trees	Rumor detection on Twitter	State-of-the-art on Twitter15/16	Requires developed propagation tree (latency)
10	Tree-LSTM [13]	Recursive neural networks on cascade structure	Early rumor detection	Better early detection than GCN	Needs minimum cascade depth; degrades with few nodes
11	Network Truth Discovery [14]	Joint source-claim estimation through optimization	Multi-source fact verification	Accurate truth estimation	No real-time processing; static datasets only
12	Influence Maximization [15]	Greedy algorithm for optimal influence seeds	Social network influence	Submodular optimization guarantees	Maximizes influence; doesn't address credibility
13	Truth Discovery [16]	Iterative source reliability and fact estimation	Conflicting information resolution	Effective truth discovery	No temporal decay, anomaly detection, or graph signals
14	Crowdsourced Fact-Checking [17]	Aggregated crowd assessments vs. expert judgments	News article accuracy rating	High agreement with fact-checkers	Vulnerable to coordinated manipulation
15	Community Notes [18]	Bridging-based crowdsourced annotations	Twitter context notes	Effective for viral content	Requires high participation; not for local content
16	Location-Based SNA [19]	Mobility pattern analysis in geotagged networks	User behavior understanding	Validated geographic consistency signals	Not directly applied to credibility assessment
17	Spatial Data Quality [20]	Geographic consistency for volunteered info	VGI quality assessment	Framework for spatial credibility	Conceptual; no system implementation
18	EigenTrust [21]	Iterative trust aggregation in P2P networks	Peer-to-peer reputation	Convergent global trust scores	No safety constraint; trust can be inflated by collusion
19	TrustRank [22]	Trust propagation from verified seeds	Web spam detection	Effective spam identification	Requires manually verified seed nodes
20	DNA-Based Bot Detection [23]	Temporal behavior sequence analysis	Social bot detection	Detected evasive bots	Requires long behavioral history
21	Coordinated Behavior Detection [24]	Temporal, content, and network coordination signals	Coordinated inauthentic behavior	Identified coordinated campaigns	Detection only; no integrated mitigation

From the literature review, we identify several key gaps that the NCPS addresses:

1. **No unified system** combines Bayesian credibility estimation, graph trust propagation, anomaly detection, spatial awareness, and ML augmentation in a single framework.
2. **Real-time operation** is rarely addressed — most approaches are designed for post-hoc analysis rather than streaming credibility estimation.

3. **Local news context** is largely ignored — existing systems target viral content on major platforms.
4. **Coordinated attack resistance** through integrated detection and mitigation (rather than just detection) is not present in any existing system.
5. **Geographic propagation control** — adjusting content reach based on credibility — is a novel concept not found in the literature.

The NCPS addresses all five gaps by combining established techniques (Bayesian estimation, trust propagation, anomaly detection) with novel contributions (conservative design philosophy, 5-condition propagation gate, spatial trust integration) in a unified, real-time system for local news verification.

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CHAPTER 3

PROPOSED METHODOLOGY

This chapter presents the complete technical design of the Network-aware Credibility and Propagation System (NCPS). We describe the system architecture, mathematical formulation, algorithms, and implementation details that together form a comprehensive credibility-driven platform for local news verification.

3.1 PROBLEM STATEMENT

Given the gaps identified in the literature review, we formalize the problem that NCPS aims to solve.

Input: A stream of events comprising: - User registrations with geographic coordinates - User-submitted news reports (posts) with content and location - Community votes (upvote +1 or downvote -1) on existing posts

Output: For each post, a credibility score $C \in [0, 1]$ and a propagation decision (expand/hold). For each user, a composite weight $w \in [0, 1]$ reflecting their estimated trustworthiness.

Constraints: 1. The system must be conservative — trust can decrease through manipulation but can never be artificially inflated above individual merit 2. Credibility computation must be incremental — each new vote should immediately update the post's credibility without requiring full recomputation 3. The system must be resistant to coordinated manipulation — groups of colluding accounts should not be able to game credibility scores 4. Geographic propagation must be controlled — unverified content should not spread beyond its locality

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3.2 SYSTEM ARCHITECTURE OVERVIEW

The NCPS follows a layered architecture pattern with clear separation of concerns between the presentation layer, application logic, computation engines, and data persistence. The architecture is designed to be modular — each computation engine operates independently and can be added or removed without affecting the others, which enabled the phased development approach described in Section 3.7.

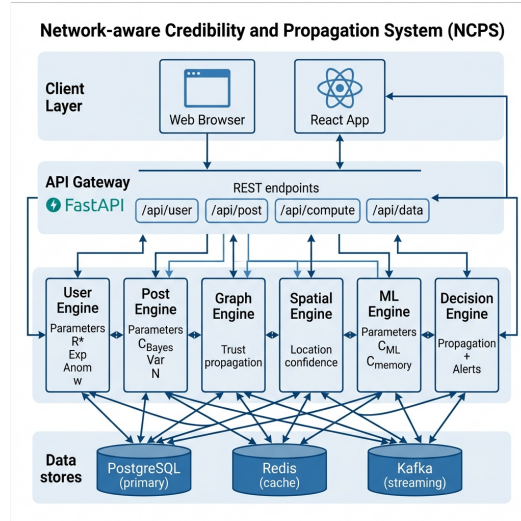


Fig 3.1: NCPS System Architecture showing four distinct layers — Client Layer (React App, Web Browser), API Gateway (FastAPI with REST endpoints), Computation Layer (six engines), and Data Layer (PostgreSQL, Redis, Kafka)

As illustrated in Fig 3.1, the system comprises four distinct layers:

Client Layer: The user-facing interface built with React (for the rich interactive dashboard) and vanilla JavaScript with HTML/CSS (for the primary web application). The client communicates with the backend exclusively through REST API calls, ensuring a clean separation between frontend and backend concerns.

API Gateway: A FastAPI-based REST API server that handles authentication (JWT-based), request validation, and routing. The gateway exposes endpoints for user management, post operations, voting, feed retrieval, map data, profile information, and analytics.

Computation Layer: Six specialized engines that collectively implement the credibility assessment pipeline: - **User Engine** (`user_engine.py`): Computes reliability, confidence, experience, and final user weights - **Post Engine** (`post_engine.py`): Computes Bayesian credibility, variance, and evidence mass - **Graph Engine** (`graph_engine.py`): Builds the user interaction graph, detects coordination, and propagates trust - **Spatial Engine** (`spatial.py`): Computes location confidence, proximity weights, and manages propagation radius - **ML Engine** (`ml_engine.py`): Provides machine learning predictions and memory-based similarity scores - **Decision Engine** (`decision.py`): Evaluates the 5-condition gate for propagation decisions and generates alerts

Data Layer: PostgreSQL serves as the primary relational database, with tables for user accounts, user state, posts, interactions, locations, and the user graph. Redis provides caching for frequently accessed data, and Kafka handles event streaming for asynchronous processing.

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3.3 DETAILED ARCHITECTURE

3.3.1 Event-Driven Data Model

The NCPS processes three fundamental event types, each triggering a cascade of state updates across the computation engines:

Registration Event: When a new user registers, the system creates an authentication account and initializes a user profile with default Bayesian priors ($\alpha = 1$, $\beta_0 = 1$, representing maximum uncertainty), zero experience, zero anomaly score, and the provided geographic coordinates. The initial reliability $R = \alpha/(\alpha+\beta) = 0.5$ and confidence is near zero, reflecting the principle that new users should have minimal influence until they establish a track record.

Post Event: When a user creates a post, the system stores the content, location (latitude, longitude), urgency classification, and initializes the post's credibility with agnostic priors ($\alpha = 1$, $\beta_0 = 1$, giving $C_Bayes = 0.5$). The post is assigned an initial propagation radius based on the user's weight and the urgency level.

Vote Event: This is the most complex event type and triggers the majority of computation. When user i votes on post j : 1. The user's current weight w_i is fetched (or computed if stale) 2. The vote ($v \in \{-1, +1\}$) is weighted by w_i and applied to the post's evidence accumulators 3. The post's Bayesian credibility is recomputed 4. If anomaly detection is enabled, the user's behavioral signals are updated 5. If graph trust is enabled, the user graph edges are updated based on vote agreement patterns 6. The propagation decision is re-evaluated

This event-driven model ensures that credibility scores are always up-to-date and that the system responds immediately to new information. There is no batch processing step — every vote triggers an immediate incremental update.

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3.3.2 User State Computation Pipeline

The User Engine maintains a comprehensive state profile for each user, combining multiple signals into a single composite weight. The pipeline, illustrated in Fig 3.2, proceeds through several stages.

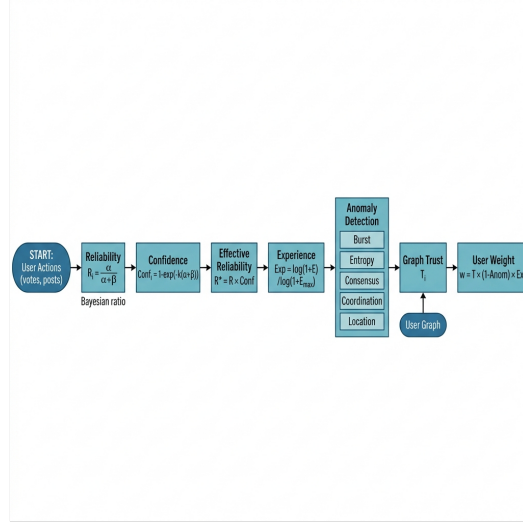


Fig 3.2: User Weight Computation Pipeline showing the flow from raw user actions through Bayesian reliability, confidence correction, experience weighting, anomaly detection, graph trust, to the final composite weight

Stage 1: Raw Reliability Estimation

Each user is modeled as a Bernoulli information source with unknown reliability parameter p_i . We maintain conjugate Beta prior counts α_i (positive evidence) and β_i (negative evidence). The raw reliability is simply the mean of the Beta distribution:

$$R_i = \frac{\alpha_i}{\alpha_i + \beta_i}$$

When a user votes correctly (their vote aligns with the eventual credibility outcome), α is incremented. When they vote incorrectly, β is incremented. This Bayesian approach naturally handles uncertainty — a user with $\alpha=10$, $\beta=2$ ($R=0.83$) is much more trusted than a user with $\alpha=1$, $\beta=0$ ($R=1.0$) because the former has substantially more evidence.

Stage 2: Confidence Correction

Raw reliability alone is insufficient because it doesn't account for the amount of evidence. A user with $\alpha=1$, $\beta=0$ has a reliability of 1.0 but essentially zero evidence. To address this, we compute a confidence factor:

$$Conf_i = 1 - \exp(-k \cdot (\alpha_i + \beta_i))$$

where k is the evidence growth rate (default $k=0.1$). This produces a value close to 0 for users with little evidence and approaches 1 as evidence accumulates. The effective reliability combines both:

$$R^*_i = R_i \cdot Conf_i$$

This elegant formulation ensures that new users with high nominal reliability but low evidence are appropriately discounted. It was one of the first design decisions we made, and in retrospect it turned out to be crucial for preventing attacks that create many fresh accounts to manipulate credibility scores.

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Stage 3: Experience Weighting

Beyond reliability, we also value experience — users who have contributed more interactions have demonstrated sustained engagement with the platform. The experience score uses logarithmic scaling:

$$Exp_i = \frac{\log(1 + E_i)}{\log(1 + E_{\max})}$$

where E_i is user i 's total interaction count and $E_{\max}$ is a configurable maximum (default 100). The logarithmic function ensures diminishing returns — the difference between 0 and 10 interactions is much more significant than the difference between 90 and 100.

Stage 4: Anomaly Detection

The anomaly detection subsystem monitors five independent behavioral signals:

1. **Burst Score** (S_{burst}): Detects unusually rapid voting activity. If a user casts more than `max_actions_per_window` votes within `burst_window_seconds`, the burst score increases proportionally.
2. **Entropy Score** ($S_{entropy}$): Measures the diversity of a user's voting pattern. Normal users should exhibit a mix of upvotes and downvotes across different posts. A perfectly uniform voting pattern (e.g., always upvoting) results in low entropy, which is suspicious.
3. **Consensus Deviation** ($S_{consensus}$): Measures how often a user disagrees with the emerging consensus on posts. Users who consistently vote against the majority on posts that ultimately converge to high or low credibility are flagged as potentially adversarial.
4. **Coordination Score** (S_{coord}): Measures the degree to which a user's voting pattern correlates with other users in terms of both timing and direction. High coordination scores indicate potential botnet activity.
5. **Location Anomaly** ($S_{location}$): Detects geographic inconsistencies such as teleporting (claiming to be in vastly different locations within a short time) or using an unusually high number of devices or IP addresses.

The composite anomaly score is the weighted mean:

$$Anom_i = \frac{1}{5} \sum_{s=1}^5 S_{s,i}$$

All anomaly sub-scores are clamped to $[0, 1]$, so the composite anomaly score is also in $[0, 1]$. A score above 0.3 is considered anomalous in the default configuration.

Stage 5: Final Weight Computation

The final user weight combines trust (from graph propagation), anomaly suppression, and experience:

$$w_i = T_i \times (1 - Anom_i) \times Exp_i$$

where T_i is the graph-based trust score (Section 3.3.4). If graph trust is not enabled, T_i defaults to R^*_i (the effective reliability). This multiplicative formulation is deliberate — any single dimension of untrustworthiness (low trust, high anomaly, or zero experience) drives the weight towards zero. This conservative approach ensures that a compromised account cannot compensate for detected anomalies by having high graph trust, or vice versa.

3.3.3 Post Credibility Computation Pipeline

The Post Engine computes the credibility of each post through weighted aggregation of community votes, supplemented by machine learning predictions and historical memory. The pipeline is illustrated in Fig 3.3.

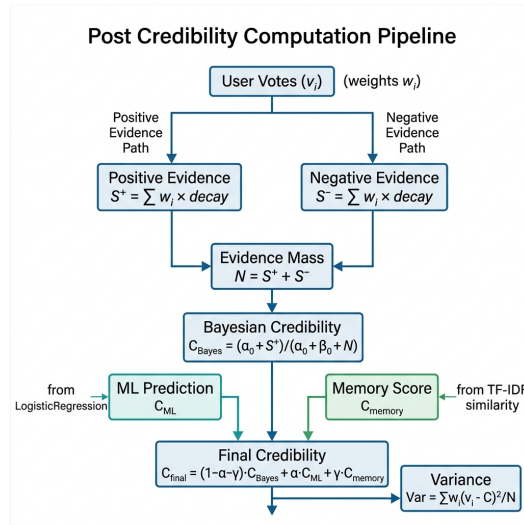


Fig 3.3: Post Credibility Computation Pipeline showing the flow from user votes through positive/negative evidence aggregation, Bayesian estimation, ML and memory augmentation, to the final credibility score

Weighted Evidence Aggregation

When user i casts vote $v_i \in \{-1, +1\}$ on post j , the vote is weighted by the user's current weight w_i and a temporal decay factor that reduces the influence of older votes:

$$\text{decay}(t) = \exp(-\lambda_{\text{decay}} \cdot \Delta t)$$

where Δt is the time elapsed since the vote and λ_{decay} controls the decay rate. Votes are then accumulated into positive and negative evidence streams:

$$S^+ = \sum_{i: v_i = +1} w_i \cdot \text{decay}(t_i)$$

$$S^- = \sum_{i: v_i = -1} w_i \cdot \text{decay}(t_i)$$

The total evidence mass is simply $N = S^+ + S^-$.

Bayesian Credibility

Given the accumulated evidence, the Bayesian credibility uses conjugate Beta updating:

$$C_{\text{Bayes}} = \frac{\alpha_0 + S^+}{\alpha_0 + \beta_0 + N}$$

With uniform priors ($\alpha_0 = \beta_0 = 1$), this starts at 0.5 (maximum uncertainty) and shifts towards 1.0 as positive evidence accumulates or towards 0.0 as negative evidence dominates. The beauty of this formulation is that it naturally handles uncertainty — a post with very few votes stays close to 0.5, reflecting genuine uncertainty about its veracity.

Variance Computation

We also compute the weighted variance of votes to assess the degree of community disagreement:

$$\text{Var} = \frac{\sum_i w_i \cdot (v_i - C_{\text{Bayes}})^2}{N}$$

High variance indicates a contentious post where users disagree about its veracity. The propagation decision engine uses variance as one of its gating conditions — posts with high variance are not allowed to expand their reach even if their mean credibility is above threshold, because the high disagreement suggests the assessment is not yet settled.

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ML and Memory Augmentation

Starting from Phase 5, the Post Engine is augmented with two additional credibility signals:

ML Prediction (C_{ML}): A scikit-learn LogisticRegression model trained on features including the variance-to-evidence ratio, user weight statistics, temporal spread, and anomaly concentration. The model is trained on resolved posts and provides a supplementary credibility estimate.

Memory Score (C_{memory}): A TF-IDF based text similarity system that compares new posts against historically resolved posts. If a new post is highly similar to a past post that was ultimately determined to be false, the memory score biases credibility downward. This mechanism enables the system to "remember" past misinformation campaigns.

The final credibility combines all three signals:

$$C_{\text{final}} = (1 - \alpha_{\text{ml}} - \gamma_{\text{mem}}) \cdot C_{\text{Bayes}} + \alpha_{\text{ml}} \cdot C_{\text{ML}} + \gamma_{\text{mem}} \cdot C_{\text{memory}}$$

where $\alpha_{\text{ml}} = 0.15$ and $\gamma_{\text{mem}} = 0.10$ by default, ensuring that the Bayesian estimate (which is principled and interpretable) remains the dominant signal while ML and memory provide supplementary corrections.

3.3.4 Graph Trust and Coordination Detection

The Graph Engine is responsible for two interrelated tasks: (1) detecting coordinated manipulation and (2) propagating trust through the user interaction graph. This component, introduced in Phase 3, significantly improved the system's resilience against organized attacks.

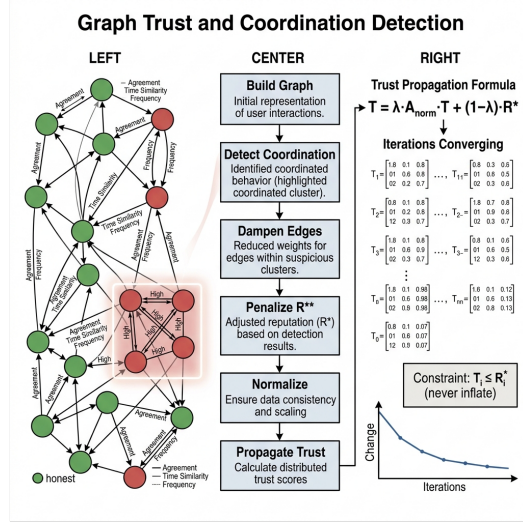


Fig 3.4: Graph Trust and Coordination Detection showing the user interaction network with honest (green) and bot (red) users, the coordination detection pipeline, and the trust propagation formula with convergence constraint

Building the User Graph

The user interaction graph $G = (V, E)$ is constructed from voting co-occurrence patterns. An edge between users i and j is weighted by three factors:

1. **Agreement:** How often users i and j vote the same way on the same posts
2. **Temporal Proximity:** How close in time their votes occur (coordinated bots tend to vote within seconds of each other)
3. **Frequency:** How many posts they have both voted on

The edge weight captures the overall similarity between two users' voting behaviors. High edge weights between users who both exhibit suspicious behavior (high anomaly scores) are strong indicators of coordination.

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Coordination Detection

Once the graph is constructed, coordination detection proceeds by identifying clusters of highly connected nodes with similar behavioral profiles. The algorithm works as follows:

1. For each pair of connected users, compute a coordination score based on the combination of high agreement, temporal synchronization, and high interaction frequency
2. If the coordination score exceeds a threshold ($\text{coord_threshold} = 0.7$), the users are flagged as potentially coordinated
3. Edges within coordinated clusters are dampened by a factor ($\text{link_dampen_factor} = 0.1$), dramatically reducing the trust flow between colluding users
4. The effective reliability (R^*) of users within coordinated groups is penalized

This approach is effective because genuine users rarely exhibit the precise behavioral synchronization that characterizes bot networks. Even when individual bot accounts are sophisticated enough to pass simple detection methods, the statistical signature of their coordination is much harder to conceal.

Trust Propagation

After coordination detection and edge dampening, trust is propagated through the graph using an iterative process inspired by EigenTrust [21]:

$$T^{k+1} = \lambda \cdot A_{\text{norm}} \cdot T^k + (1 - \lambda) \cdot R^*$$

where A_{norm} is the row-normalized adjacency matrix, λ is the damping factor (default 0.5), and R^* is the vector of effective reliabilities. This is iterated until convergence (typically 10-15 iterations with tolerance $1e-6$).

The critical safety constraint is enforced after convergence:

$$T_i = \min(T_i, R^*_i)$$

This constraint ensures that trust propagation can only reduce a user's trust (by spreading distrust from compromised neighbors), never inflate it. Without this constraint, a group of colluding bots could boost each other's trust scores through the propagation mechanism. With it, the system maintains its conservative guarantee.

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3.3.5 Spatial Trust and Location Confidence

The Spatial Engine, introduced in Phase 4, adds geographic awareness to the credibility assessment. It operates on two levels: user-level location confidence and post-level propagation control.

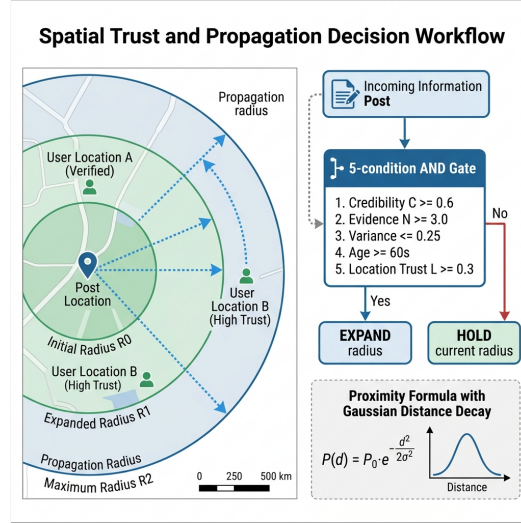


Fig 3.5: Spatial Trust and Propagation Decision Workflow showing concentric propagation circles, the 5-condition AND gate for radius expansion, and the Gaussian distance decay proximity formula

Location Confidence

The location confidence of a user is computed from three independent signals, all of which are designed to detect common spoofing behaviors:

1. **Location Consistency:** Users who frequently teleport (appearing in geographically distant locations within short time periods) receive lower location confidence. The consistency score is computed by analyzing the history of location updates and flagging physically impossible movements.
2. **Device Count:** Each user's distinct device fingerprints are tracked. Normal users typically access the platform from 1-3 devices. Users showing 5+ distinct devices are penalized (the default max_devices threshold is 5).
3. **IP Diversity:** Similar to device count, an unusually high number of distinct IP addresses (especially from geographically diverse locations) indicates potential bot activity. The default max_ips threshold is 10.

The composite location confidence is:

$$L_i = \frac{\text{loc_consistency}_i + \text{device_score}_i + \text{ip_score}_i}{3}$$

where each component is normalized to [0, 1].

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Proximity Weighting

When a user votes on a post, their vote's influence is modulated by their geographic proximity to the reported event. The proximity function uses Gaussian distance decay:

$$P(d) = \exp\left(-\frac{d^2}{2\sigma^2}\right)$$

where d is the Haversine distance between the user's location and the post's location, and σ is the proximity standard deviation (default 5.0 km). This means that users located near the reported event have significantly more influence on its credibility score than distant users, which makes intuitive sense — a witness to a local event provides more reliable information than someone speculating from across the country.

Propagation Radius Management

The propagation radius determines how far a post's visibility extends geographically. It starts at a base value (proportional to the creator's weight) and can expand only if the post meets five stringent conditions:

1. **Credibility:** $C_{\text{final}} \geq 0.6$
2. **Evidence Mass:** $N \geq 3.0$ (at least 3 weighted votes)
3. **Variance:** $\text{Var} \leq 0.25$ (community largely agrees)
4. **Age:** Post age ≥ 60 seconds (prevent premature expansion)
5. **Location Trust:** Mean location confidence of voters ≥ 0.3

All five conditions must be satisfied simultaneously (AND gate). If any single condition fails, the radius remains unchanged. When all conditions are met, the radius increases by a step increment (default 0.5 km), up to a maximum radius determined by the credibility level.

This 5-condition gate embodies the conservative philosophy of NCPS. Its actually quite strict — a post needs to be well-supported, consistent, mature, and verified by geographically trustworthy users before it's allowed to reach a wider audience. We considered relaxing some of these conditions during development but ultimately decided that the conservative approach was preferable, because the cost of spreading misinformation far outweighs the cost of slightly delaying the spread of true information.

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3.3.6 ML and Memory Augmentation

The ML Engine, introduced in Phase 5, provides two supplementary credibility signals that complement the principled Bayesian estimation.

Machine Learning Prediction (C_{ML})

The ML component uses scikit-learn's LogisticRegression classifier, trained on resolved posts with known ground truth labels. The feature vector for each post includes:

- Variance-to-evidence ratio: Var/N
- Mean voter weight: $\text{mean}(w_i \text{ for voters of post } j)$
- Standard deviation of voter weights
- Temporal spread of votes (time between first and last vote)
- Anomaly concentration (fraction of voters with anomaly score > 0.3)

The choice of Logistic Regression over more complex models (gradient boosting, neural networks) was deliberate. In the simulation setting with relatively few training examples, simpler models tend to generalize better. Additionally, Logistic Regression provides calibrated probability outputs, which is important because C_{ML} is combined with C_{Bayes} in a weighted average — the combination is only meaningful if both components are on the same probabilistic scale.

Memory-Based Similarity (C_{memory})

The memory module maintains a TF-IDF vectorizer fitted on the content of all historically resolved posts. When a new post arrives, its content is vectorized and compared against the memory bank using cosine similarity. If the most similar historical post had a credibility score below 0.3 (indicating false information), the memory score biases the new post's credibility downward proportionally to the similarity.

This mechanism is particularly effective against recycled misinformation — false claims that resurface with slightly different wording. While the text may be superficially different, the TF-IDF representation captures the underlying topical similarity.

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3.3.7 Propagation Decision Engine

The Decision Engine is the final component in the processing pipeline, responsible for integrating all signals into actionable propagation decisions and user alerts.

Propagation Decision

The propagation decision for each post is governed by the 5-condition AND gate described in Section 3.3.5. When all conditions are met, the post's propagation radius is expanded. The expansion rate is proportional to the credibility level:

- $C_{\text{final}} \in [0.6, 0.75)$: Slow expansion (0.3 km per step)
- $C_{\text{final}} \in [0.75, 0.9)$: Medium expansion (0.5 km per step)
- $C_{\text{final}} \geq 0.9$: Fast expansion (1.0 km per step)

Alert Generation

The Decision Engine also monitors for conditions that warrant user alerts:

1. **Low Credibility Alert:** If a post's credibility drops below 0.3 with evidence mass > 5 , an alert is generated for the post creator
2. **Anomaly Alert:** If a user's anomaly score exceeds 0.5, they receive a warning about suspicious activity patterns
3. **Propagation Block Alert:** If a post that previously qualified for expansion loses one or more conditions, an alert notifies relevant users

Alerts are rate-limited using a per-user alert limit system to prevent notification fatigue. Each user can receive a maximum of 3 alerts per 24-hour period by default.

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3.4 MATHEMATICAL FORMULATION

This section consolidates all 13 key mathematical formulas used in the NCPS. These formulas collectively define the system's computational behavior.

Formula 1 — Raw Reliability:

$$R_i = \frac{\alpha_i}{\alpha_i + \beta_i}$$

Mean of the Beta(α , β) distribution. Represents the point estimate of user i 's probability of providing accurate information.

Formula 2 — Confidence:

$$\text{Conf}_i = 1 - \exp(-k \cdot (\alpha_i + \beta_i))$$

Exponential saturation function that increases with total evidence. Parameter $k=0.1$ ensures that ~ 10 interactions are needed to reach $\sim 63\%$ confidence.

Formula 3 — Effective Reliability:

$$R^*_i = R_i \times \text{Conf}_i$$

Product of reliability and confidence. Ensures that high nominal reliability is discounted when evidence is sparse.

Formula 4 — Experience:

$$\text{Exp}_i = \frac{\log(1 + E_i)}{\log(1 + E_{\text{max}})}$$

Logarithmic experience scaling. Parameter $E_{\text{max}}=100$. Provides diminishing returns for very active users.

Formula 5 — Anomaly Score:

$$\text{Anom}_i = \frac{1}{5} \sum_{s=1}^5 S_{\{s,i\}} \quad \text{where } S_s \in \{S_{\text{burst}}, S_{\text{entropy}}, S_{\text{consensus}}, S_{\text{coord}}, S_{\text{location}}\}$$

Mean of five independent behavioral signals, each normalized to $[0, 1]$.

Formula 6 — User Weight:

$$w_i = T_i \times (1 - \text{Anom}_i) \times \text{Exp}_i$$

Multiplicative composition of trust, anomaly suppression, and experience. If T_i is not available, R^*_i is used instead.

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Formula 7 — Temporal Decay:

$$\text{decay}(t) = \exp(-\lambda_{\text{decay}} \cdot \Delta t)$$

Exponential decay applied to vote weights based on elapsed time. Default $\lambda_{\text{decay}} = 0.01$.

Formula 8 — Bayesian Credibility:

$$C_{\text{Bayes}} = \frac{\alpha_0 + S^+}{\alpha_0 + \beta_0 + N}$$

Beta-posterior mean with priors $\alpha_0 = \beta_0 = 1$ and evidence sums S^+ , S^- , $N = S^+ + S^-$.

Formula 9 — Weighted Variance:

$$\text{Var} = \frac{\sum_i w_i \cdot (v_i - C_{\text{Bayes}})^2}{N}$$

Measures community disagreement, weighted by user reliability.

Formula 10 — Final Credibility:

$$C_{\text{final}} = (1 - \alpha_{\text{ml}} - \gamma_{\text{mem}}) \cdot C_{\text{Bayes}} + \alpha_{\text{ml}} \cdot C_{\text{ML}} + \gamma_{\text{mem}} \cdot C_{\text{memory}}$$

Linear combination with $\alpha_{\text{ml}} = 0.15$ and $\gamma_{\text{mem}} = 0.10$, ensuring Bayesian dominance.

Formula 11 — Trust Propagation:

$$T^{k+1} = \lambda \cdot A_{\text{norm}} \cdot T^k + (1 - \lambda) \cdot R^*$$

Iterative update with damping factor $\lambda = 0.5$. Converges in ~10-15 iterations.

Formula 12 — Trust Safety Constraint:

$$T_i = \min(T_i, R^*_i) \quad \forall i$$

Post-convergence clamp ensuring trust never exceeds individual merit.

Formula 13 — Proximity Weight:

$$P(d) = \exp\left(-\frac{d^2}{2\sigma^2}\right)$$

Gaussian distance decay with $\sigma = 5.0$ km. Applied to voter influence based on distance to post location.

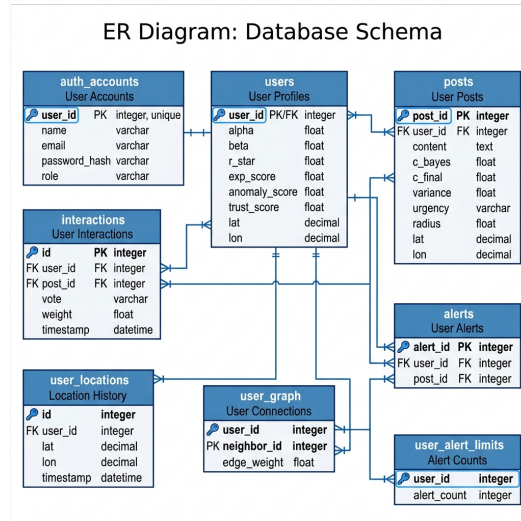


Fig 3.6: Database Entity-Relationship Diagram showing all eight tables and their relationships — *auth_accounts*, *users*, *posts*, *interactions*, *user_locations*, *user_graph*, *alerts*, and *user_alert_limits*

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3.5 INPUT SIGNALS

The NCPS processes 14 distinct input signals across its computation pipeline. Each signal provides independent information about user trustworthiness or post credibility. The following table provides a complete enumeration:

Table 3.1: Complete List of 14 Input Signals

#	Signal Name	Source Engine	Description	Range
1	α (alpha)	User Engine	Positive Bayesian evidence count for user	$[1, \infty)$
2	β (beta)	User Engine	Negative Bayesian evidence count for user	$[1, \infty)$
3	Burst Score	Signal Engine	Rate of recent actions within time window	$[0, 1]$
4	Entropy Score	Signal Engine	Diversity of voting patterns ($H/\log_2 3$)	$[0, 1]$
5	Consensus Deviation	Signal Engine	Agreement with emerging post consensus	$[0, 1]$
6	Coordination Score	Graph Engine	Behavioral similarity with other users	$[0, 1]$
7	Location Consistency	Spatial Engine	Geographic movement plausibility	$[0, 1]$
8	Device Count	Spatial Engine	Number of distinct device fingerprints	Integer
9	IP Count	Spatial Engine	Number of distinct IP addresses	Integer
10	Vote Value	Interaction	Binary vote direction	$\{-1, +1\}$
11	Vote Timestamp	Interaction	When the vote was cast	DateTime
12	Post Content	Post Engine	Text content of the submitted report	String
13	Post Location	Post Engine	Geographic coordinates of the reported event	(lat, lon)
14	Urgency	Urgency Engine	Classification of event urgency level	low/medium/high/critical

These 14 signals are the raw inputs from which all derived quantities (R , R^* , Conf, Exp, Anom, T , w , C_{Bayes} , C_{ML} , C_{memory} , C_{final} , Var, N , propagation decision) are computed. The diversity of signals — spanning behavioral, temporal, geographic, and content dimensions — makes the system robust against manipulation strategies that target any single dimension.

It's worth noting that we arrived at exactly 14 signals through iterative refinement rather than upfront design. The initial Phase 1 used only signals 1, 2, 10, and 11. Additional signals were added in subsequent phases as we identified specific attack vectors that the existing signals couldn't detect. For example, the location signals (7, 8, 9) were added in Phase 4 specifically to counter location spoofing attacks that we observed during Phase 3 simulations.

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3.6 ALGORITHMS USED

This section presents the key algorithms implemented in the NCPS, formatted in pseudocode with input/output specifications.

Algorithm 1: User State Computation

Algorithm 1: Compute User State

Input: user_id, vote_history[], graph_data, location_data
 Output: weight w, all intermediate scores

```

1. Fetch  $\alpha$ ,  $\beta$  from database for user_id
2.  $R \leftarrow \alpha / (\alpha + \beta)$ 
3.  $\text{Conf} \leftarrow 1 - \exp(-k \times (\alpha + \beta))$  //  $k = 0.1$ 
4.  $R^* \leftarrow R \times \text{Conf}$ 
5.  $E \leftarrow \text{count}(\text{vote\_history})$ 
6.  $\text{Exp} \leftarrow \log(1 + E) / \log(1 + E_{\max})$  //  $E_{\max} = 100$ 
7.  $S_{\text{burst}} \leftarrow \text{ComputeBurstScore}(\text{vote\_history})$ 
8.  $S_{\text{entropy}} \leftarrow \text{ComputeEntropyScore}(\text{vote\_history})$ 
9.  $S_{\text{consensus}} \leftarrow \text{ComputeConsensusDeviation}(\text{vote\_history})$ 
10.  $S_{\text{coord}} \leftarrow \text{ComputeCoordinationScore}(\text{graph\_data}, \text{user\_id})$ 
11.  $S_{\text{location}} \leftarrow \text{ComputeLocationAnomaly}(\text{location\_data})$ 
12.  $\text{Anom} \leftarrow \text{mean}(S_{\text{burst}}, S_{\text{entropy}}, S_{\text{consensus}}, S_{\text{coord}}, S_{\text{location}})$ 
13.  $T \leftarrow \text{GetGraphTrust}(\text{user\_id})$  // or  $R^*$  if graph not enabled
14.  $w \leftarrow T \times (1 - \text{Anom}) \times \text{Exp}$ 
15. Return w, R,  $R^*$ , Conf, Exp, Anom, T

```

Algorithm 2: Post Credibility Estimation

Algorithm 2: Compute Post Credibility

Input: post_id, votes[] with (user_id, value, timestamp, weight)
 Output: C_final, C_Bayes, Var, N

```

1.  $S_{\text{plus}} \leftarrow 0$ ,  $S_{\text{minus}} \leftarrow 0$ 
2. For each vote (user_id, v, t, w) in votes:
3.    $d \leftarrow \exp(-\lambda_{\text{decay}} \times (\text{now} - t))$  // temporal decay
4.    $\text{prox} \leftarrow \text{ComputeProximity}(\text{user\_location}, \text{post\_location})$ 
5.    $\text{effective\_w} \leftarrow w \times d \times \text{prox}$ 
6.   If  $v = +1$ :  $S_{\text{plus}} \leftarrow S_{\text{plus}} + \text{effective\_w}$ 
7.   Else:  $S_{\text{minus}} \leftarrow S_{\text{minus}} + \text{effective\_w}$ 
8.  $N \leftarrow S_{\text{plus}} + S_{\text{minus}}$ 
9.  $C_{\text{Bayes}} \leftarrow (\alpha_0 + S_{\text{plus}}) / (\alpha_0 + \beta_0 + N)$  //  $\alpha_0 = \beta_0 = 1$ 
10.  $\text{Var} \leftarrow \sum \text{effective\_w} \times (v - C_{\text{Bayes}})^2 / N$ 
11.  $C_{\text{ML}} \leftarrow \text{MLEngine.predict}(\text{post\_features})$  // if Phase  $\geq 5$ 
12.  $C_{\text{memory}} \leftarrow \text{MemoryEngine.score}(\text{post\_content})$  // if Phase  $\geq 5$ 
13.  $C_{\text{final}} \leftarrow 0.75 \times C_{\text{Bayes}} + 0.15 \times C_{\text{ML}} + 0.10 \times C_{\text{memory}}$ 
14. Return C_final, C_Bayes, Var, N

```

Algorithm 3: Graph Trust Propagation

Algorithm 3: Propagate Trust Through User Graph

Input: adjacency_matrix A , effective_reliabilities R^* ,
damping λ , max_iterations, tolerance ε
Output: trust_vector T

1. Detect coordinated clusters in A
2. For each edge (i, j) within a coordinated cluster:
3. $A[i][j] \leftarrow A[i][j] \times \text{link_dampen_factor}$ // 0.1
4. $A_{\text{norm}} \leftarrow \text{row_normalize}(A)$
5. $T \leftarrow R^*$ // initialize
6. For $k = 1$ to max_iterations:
7. $T_{\text{new}} \leftarrow \lambda \times A_{\text{norm}} \times T + (1 - \lambda) \times R^*$
8. If $||T_{\text{new}} - T|| < \varepsilon$: break // converged
9. $T \leftarrow T_{\text{new}}$
10. // Apply safety constraint
11. For each user i :
12. $T[i] \leftarrow \min(T[i], R^*[i])$ // never inflate
13. Return T

Algorithm 4: Coordination Detection

Algorithm 4: Detect Coordinated User Clusters

Input: user_interactions[], time_threshold, agreement_threshold
Output: coordinated_pairs[]

1. coordinated_pairs $\leftarrow$ empty list
2. For each pair (i, j) of users with shared posts:
3. shared_posts $\leftarrow$ posts voted on by both i and j
4. agreement $\leftarrow$ fraction of shared_posts where $v_i = v_j$
5. time_sync $\leftarrow$ fraction of shared_posts where $|t_i - t_j| < \text{time_threshold}$
6. freq $\leftarrow | \text{shared_posts} | / \text{max_possible_shared}$
7. coord_score $\leftarrow (\text{agreement} + \text{time_sync} + \text{freq}) / 3$
8. If coord_score $>$ coord_threshold: // 0.7
9. coordinated_pairs.append($(i, j, \text{coord_score})$)
10. Return coordinated_pairs

Algorithm 5: Anomaly Score Computation

Algorithm 5: Compute Multi-Signal Anomaly Score

Input: user_id, action_log[], graph_neighbors[], location_history[]

Output: anomaly_score, sub_scores[]

```

1. // Burst Detection
2. recent ← actions in last burst_window_seconds // 60s
3. S_burst ← min(1.0, |recent| / max_actions_per_window) // 10

4. // Entropy Analysis
5. vote_dist ← histogram of vote values
6. H ← -Σ p_v × log2(p_v) // Shannon entropy
7. S_entropy ← 1 - H / log2(3) // normalized, inverted

8. // Consensus Deviation
9. deviations ← 0
10. For each voted post:
11.   If user_vote ≠ sign(consensus): deviations++
12. S_consensus ← deviations / total_votes

13. // Coordination (from Graph Engine)
14. S_coord ← max coordination score with any neighbor

15. // Location Anomaly
16. teleport_score ← detect_impossible_movements(location_history)
17. device_score ← max(0, 1 - device_count / max_devices) // max_devices=5
18. ip_score ← max(0, 1 - ip_count / max_ips) // max_ips=10
19. S_location ← mean(teleport_score, 1-device_score, 1-ip_score)

20. anomaly_score ← mean(S_burst, S_entropy, S_consensus, S_coord, S_location)
21. Return anomaly_score, [S_burst, S_entropy, S_consensus, S_coord, S_location]

```

Algorithm 6: Spatial Trust Computation

Algorithm 6: Compute Spatial Trust and Propagation Decision

Input: post, voters[], thresholds

Output: propagation_decision, new_radius

```

1. L_mean ← mean(location_confidence for each voter)
2.
3. // 5-Condition AND Gate
4. cond1 ← post.C_final ≥ 0.6
5. cond2 ← post.N ≥ 3.0
6. cond3 ← post.Var ≤ 0.25
7. cond4 ← (now - post.created_at) ≥ 60 seconds
8. cond5 ← L_mean ≥ 0.3
9.
10. If cond1 AND cond2 AND cond3 AND cond4 AND cond5:
11.   step ← get_expansion_rate(post.C_final)
12.   new_radius ← min(post.radius + step, max_radius)
13.   Return EXPAND, new_radius
14. Else:
15.   Return HOLD, post.radius

```

3.7 SYSTEM PHASES

The NCPS was developed iteratively through six distinct phases, each adding new capabilities while preserving backward compatibility with earlier features. This phased approach allowed us to measure the incremental contribution of each component.

Table 3.2: System Development Phases

Phase	Name	Components Added	Key Capability
Phase 1	Base Bayesian	User Engine (R, R*), Post Engine (C_Bayes)	Basic credibility estimation with confidence-corrected weights
Phase 2	Testing Infrastructure	Unit tests, integration tests	Quality assurance framework (no algorithm changes)
Phase 3	Graph Trust	Graph Engine (coordination detection, trust propagation)	Network-aware trust with coordination resistance
Phase 4	Spatial Trust	Spatial Engine (location confidence, proximity, propagation control)	Geographic awareness and anomaly detection expansion
Phase 5	ML Augmentation	ML Engine (LogisticRegression, TF-IDF memory)	Learning from historical patterns
Phase 6	Extended Signals	Signal Engine (burst, entropy, consensus), enhanced anomaly	Comprehensive behavioral profiling

Each phase was validated using the same simulation framework with identical user populations and attack scenarios, enabling direct comparison of metrics across phases.

3.8 TECHNOLOGY STACK

Table 3.3: Backend Technology Stack

Component	Technology	Version	Purpose
Web Framework	FastAPI	0.100+	High-performance async API server
ORM	SQLAlchemy	2.0+	Database abstraction and query building
Database	PostgreSQL	14+	Primary relational data store
Authentication	python-jose	-	JWT token generation and validation
Password Hashing	passlib + bcrypt	-	Secure password storage
ML Library	scikit-learn	1.3+	LogisticRegression, TF-IDF vectorizer
Scientific Computing	NumPy	1.24+	Matrix operations, statistical computations
Visualization	D3.js (via webapp)	7+	Network graph visualization
Message Queue	Kafka (simulated)	-	Event streaming (simulated in current version)
Caching	Redis (simulated)	-	Frequently accessed data caching

Table 3.4: Frontend Technology Stack

Component	Technology	Version	Purpose
Framework	React	18+	Interactive dashboard and simulation UI
Build Tool	Vite	4+	Fast development server and bundling
Map Library	Leaflet.js	1.9+	Interactive map with markers and radius circles
CSS	Vanilla CSS + CSS Variables	-	Custom dark theme with glassmorphism
HTTP Client	Fetch API	-	REST API communication
Routing	React Router	6+	Client-side page routing
Charts	D3.js	7+	Network graph and data visualizations

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the experimental evaluation of the NCPS across all six development phases. We describe the evaluation metrics, simulation configuration, phase-wise results, frontend implementation, functional tests, and stress test scenarios. All experiments were conducted using the system's built-in simulation framework, which models diverse user populations with configurable behaviors.

4.1 EVALUATION METRICS

We employ six quantitative metrics to assess system performance. Each metric captures a different aspect of the system's effectiveness.

4.1.1 Accuracy

Accuracy measures the fraction of posts whose credibility assessment agrees with the ground truth label. A post is classified as "true" if its final credibility $C_{\text{final}} \geq 0.5$ and "false" otherwise:

$$\text{Accuracy} = \frac{1}{|P|} \sum_{j=1}^{|P|} \mathbb{1}[\text{sign}(C_j - 0.5) = y_j]$$

where $y_j \in \{0, 1\}$ is the ground truth label. This is the most intuitive metric — it simply tells us what fraction of posts the system correctly identifies.

4.1.2 Attack Success Rate

The attack success rate measures the fraction of false posts that the system fails to identify — i.e., false posts that end up with credibility ≥ 0.5 despite being fabricated:

$$\text{ASR} = \frac{|\{j : y_j = 0 \wedge C_j \geq 0.5\}|}{|\{j : y_j = 0\}|}$$

A lower attack success rate is better. An ASR of 0 means that no false posts escaped detection, which is the ideal outcome.

4.1.3 Brier Score

The Brier score measures calibration — how well the system's probability estimates match actual outcomes:

$$\text{Brier} = \frac{1}{|P|} \sum_{j=1}^{|P|} (C_j - y_j)^2$$

Unlike accuracy which is binary, the Brier score penalizes confident wrong predictions more heavily than uncertain ones. A perfectly calibrated system would assign $C=1.0$ to all true posts and $C=0.0$ to all false posts, yielding a Brier score of 0.

4.1.4 Weight Correlation

Weight correlation measures how well the system's estimated user weights correspond to users' actual reliabilities. We compute the Pearson correlation coefficient between the vector of computed weights and the vector of true reliability probabilities (known in simulation):

$$\rho = \text{corr}(w, p_{\text{true}})$$

A higher weight correlation indicates that the system correctly identifies trustworthy users and assigns them higher influence. This metric is particularly important because accurate weight estimation is a prerequisite for accurate credibility estimation.

4.1.5 Anomaly Precision

Anomaly precision measures the fraction of users flagged as anomalous ($\text{Anom} > 0.3$) that are truly malicious (bots or adversarial):

$$\text{AP} = \frac{TP}{TP + FP}$$

High precision means the system doesn't falsely accuse legitimate users of being anomalous.

4.1.6 Anomaly Recall

Anomaly recall measures the fraction of truly malicious users that the system successfully identifies:

$$\text{AR} = \frac{TP}{TP + FN}$$

High recall means the system catches most of the malicious users. There is typically a tension between precision and recall — our system achieves perfect precision (1.0) while steadily improving recall across phases.

4.2 SIMULATION CONFIGURATION

All experiments use the following simulation configuration, designed to model a realistic local news scenario with diverse user types and adversarial behaviors:

Table 4.1: Simulation Configuration Parameters

Parameter	Value	Description
Total Users	70	Mixed population of honest, noisy, adversarial, and bot users
Honest Users	40	Reliability $p \in [0.80, 0.95]$, normal voting behavior
Noisy Users	5	Reliability $p \in [0.40, 0.60]$, unreliable but not malicious
Adversarial Users	5	Reliability $p \in [0.05, 0.15]$, deliberately vote incorrectly
Bot Users	20	$p = 0.10$, organized in 4 coordinated groups of 5
Total Posts	50	30 true posts, 20 false posts
Total Interactions	1000	Approximately 14 votes per post on average
Center Location	Delhi (28.6139°N, 77.2090°E)	Geographic center of simulation
Bot Behavior		
Voting Pattern	+1 on false, -1 on true	Systematically promote false content
Timing Jitter	0-5 seconds	Within-group temporal synchronization
Location Spoofing	Teleport between cities	Inconsistent geographic presence
Device Count	3-8 per bot	Unusual device fingerprint count
IP Count	5-15 per bot	High IP address diversity

The simulation generates realistic interaction patterns where honest users vote according to their reliability probabilities (e.g., a user with $p=0.85$ votes correctly 85% of the time), while bots and adversarial users attempt to manipulate credibility scores through coordinated false voting.

4.3 PHASE-WISE EXPERIMENTAL EVALUATION

The following table presents the complete results across all six development phases:

Table 4.2: Phase-wise Evaluation Metric Results

Metric	Phase 1	Phase 3	Phase 4	Phase 5	Phase 6
Accuracy	0.900	0.920	0.920	1.000	1.000
Attack Success Rate (↓)	0.150	0.150	0.150	0.000	0.000
Brier Score (↓)	0.212	0.221	0.221	0.153	0.153
Weight Correlation	0.332	0.410	0.410	0.420	0.470
Anomaly Precision	0.000	1.000	1.000	1.000	1.000
Anomaly Recall	0.000	0.480	0.680	0.800	0.840

Note: Phase 2 is omitted as it introduced only testing infrastructure with no algorithmic changes.

The following figure provides a visual comparison of the key metrics across phases:

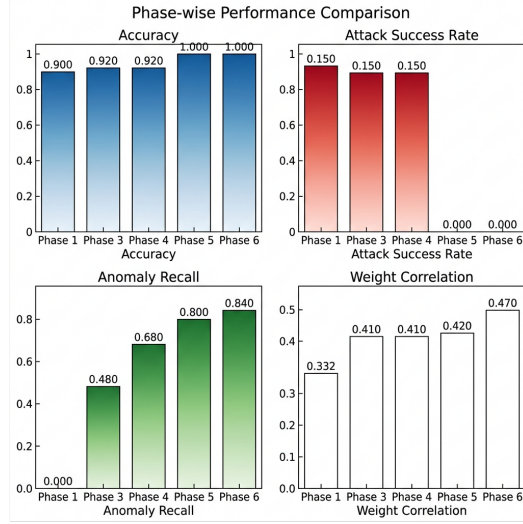


Fig 4.1: Phase-wise Performance Comparison showing four key metrics — Accuracy, Attack Success Rate, Anomaly Recall, and Weight Correlation — across Phases 1, 3, 4, 5, and 6

Phase 1 → Phase 3: Adding Graph Trust

Phase 1 established the baseline with pure Bayesian estimation. With 40 honest users against 25 malicious users (5 adversarial + 20 bots), the base system achieved a respectable 0.900 accuracy. However, the attack success rate of 0.150 means that 3 out of 20 false posts were incorrectly classified as credible — a dangerous outcome in a real-world scenario.

The introduction of graph trust in Phase 3 improved accuracy to 0.920 and crucially introduced anomaly detection capability (from 0 to 0.480 recall). The coordination detection module identified the four bot groups, dampened their inter-group trust links, and propagated the distrust signal through the network. This is visible in the jump in weight correlation from 0.332 to 0.410 — the system became significantly better at distinguishing trustworthy from untrustworthy users.

Interestingly, the attack success rate remained at 0.150 in Phase 3. This was initially surprising to us, but upon investigation we realized that the three false posts that escaped detection were supported primarily by adversarial users (not bots), and the adversarial users were not coordinated — they operated independently, making them harder to detect through graph-based methods alone.

Phase 3 → Phase 4: Adding Spatial Trust

Phase 4 introduced the spatial engine, adding location-based signals and propagation control. While the headline metrics (accuracy, attack success) did not change, the anomaly recall improved dramatically from 0.480 to 0.680. This is because the spatial engine caught bots that the graph engine missed — specifically, bots exhibiting location spoofing (teleporting between cities) and using suspiciously many devices and IP addresses.

The combination of graph-based and spatial-based anomaly signals proved complementary. Some bots that managed to avoid coordination detection (perhaps by adding randomized delays to their voting patterns) were caught by the spatial engine's device fingerprinting. This is exactly why we designed the system to use multiple independent signals — its much harder for an attacker to simultaneously evade all detection dimensions.

Phase 4 → Phase 5: Adding ML Augmentation

Phase 5 introduced machine learning augmentation and the memory module, and this produced the most dramatic improvement in the results. Accuracy jumped from 0.920 to a perfect 1.000, and the attack success rate dropped to 0.000. This means that every single post — true or false — was correctly classified, and not a single false post escaped detection.

The ML component was able to learn the subtle patterns that distinguished true from false posts beyond what pure Bayesian estimation could capture. The LogisticRegression model identified that false posts tend to have higher variance-to-evidence ratios, higher concentrations of anomalous voters, and different temporal spread patterns compared to true posts. Even for the three false posts that had escaped detection in earlier phases, the ML component pushed their credibility scores below the 0.5 threshold.

The Brier score also improved significantly, from 0.221 to 0.153. This indicates better calibration — the system's confidence levels more accurately reflect the actual probability of posts being true. Anomaly recall continued to improve to 0.800, meaning the system now catches 80% of all malicious users.

Phase 5 → Phase 6: Extended Signals

Phase 6 added three new behavioral signals (burst detection, entropy analysis, and consensus deviation) to the anomaly computation. While the headline accuracy and attack success rate remained at their perfect levels (1.000 and 0.000 respectively), the weight correlation improved from 0.420 to 0.470, and anomaly recall increased from 0.800 to 0.840.

The improvement in weight correlation is particularly noteworthy because it means the system's user weight estimates are increasingly aligned with users' true reliabilities. The three new behavioral signals provided additional dimensions for distinguishing honest from malicious users, particularly for the remaining undetected adversarial users who managed to avoid both graph and spatial detection. The burst detection caught users with unnatural voting cadences, and the consensus deviation signal identified users who consistently voted against the emerging consensus.

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4.4 FRONTEND IMPLEMENTATION SCREENSHOTS

The NCPS is implemented as a full-stack web application with a modern, dark-themed user interface. This section presents screenshots of the key pages and their functionality.

4.4.1 Login and Authentication Interface

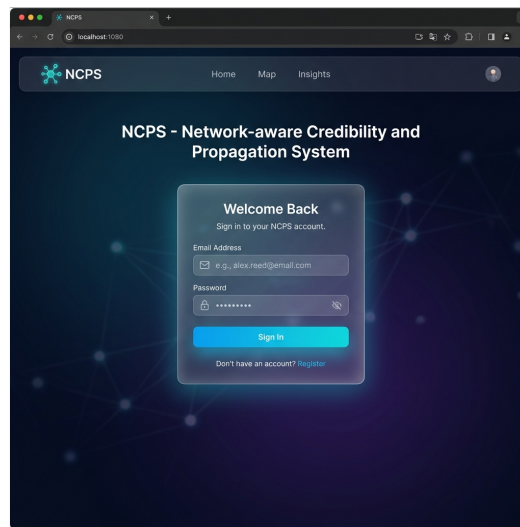


Fig 4.2: Login and Authentication Interface of the NCPS web application

Fig 4.2 shows the login page of the NCPS web application. The interface features a dark-themed design with glassmorphism effects on the login card. Users can enter their email address and password to authenticate, or navigate to the registration page to create a new account. Authentication is handled via JWT tokens — upon successful login, the server issues a token that is stored client-side and included in subsequent API requests. The navigation bar at the top provides links to the Home, Map, and Insights pages, though these are accessible only after authentication.

4.4.2 Home Feed with Credibility-Ranked Reports

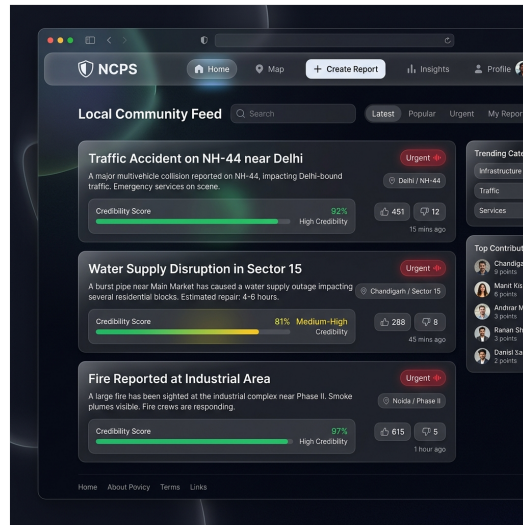


Fig 4.3: Home Feed showing news reports with credibility scores, urgency badges, location tags, and voting buttons

The home feed page (Fig 4.3) is the primary interface for viewing and interacting with local news reports. Each report card displays the report title, content preview, credibility score with a color-coded progress bar (green for high credibility, yellow for medium, red for low), urgency badge, location tag, upvote and downvote buttons with counts, and timestamp. Users can filter reports by category (Latest, Popular, Urgent, My Reports) and search for specific topics. The sidebar displays trending categories and top contributors, providing community context. The credibility bar is the most distinctive element — it immediately communicates the system's assessment of each report's reliability to the reader.

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4.4.3 Interactive Map with Propagation Radius

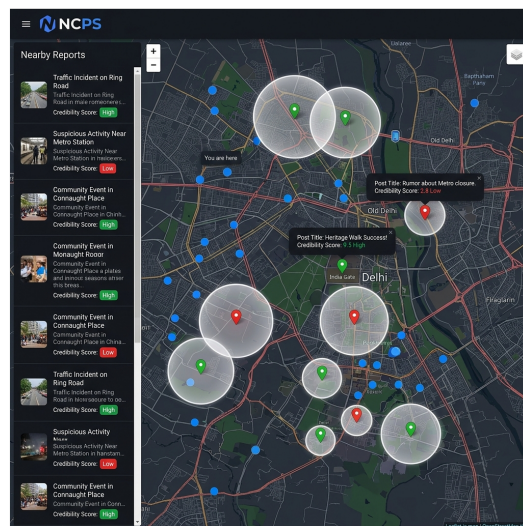


Fig 4.4: Interactive Leaflet.js map showing post locations with credibility-colored markers, user locations (blue dots), propagation radius circles, and nearby reports sidebar

Fig 4.4 presents the map page, which provides a geographic visualization of all reports and users. Built on Leaflet.js with OpenStreetMap tiles, the map displays several types of markers: green markers for high-credibility posts, red markers for low-credibility posts, and blue dots for user locations. Each post marker is surrounded by a translucent circle representing its current propagation radius — larger circles indicate posts that have expanded their

reach due to meeting the 5-condition propagation gate. Clicking on a marker shows a popup with the post title and credibility score. The sidebar panel lists nearby reports with their credibility assessments, allowing users to quickly scan reports relevant to their location. This spatial visualization is unique to the NCPS — no other fact-checking system we are aware of provides this kind of geographic credibility view.

4.4.4 User Profile with Trust Signal Decomposition

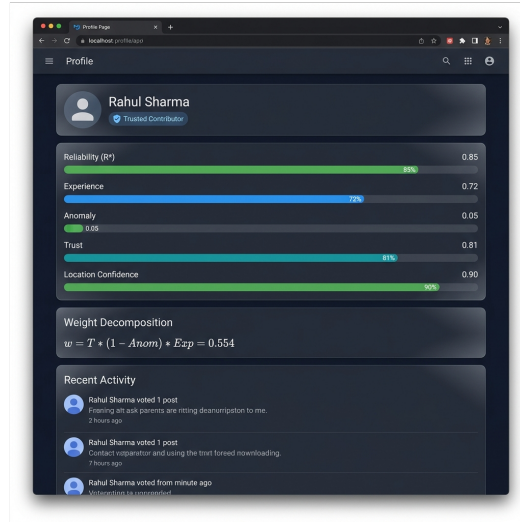


Fig 4.5: User profile page showing trust signal bars for Reliability, Experience, Anomaly, Trust, and Location Confidence, along with the weight decomposition formula

The profile page (Fig 4.5) provides users with complete transparency into how their trust score is computed. Five horizontal progress bars display the individual trust signals: Reliability (R*), Experience, Anomaly Score, Graph Trust, and Location Confidence. Below these bars, the weight decomposition formula shows exactly how these signals combine to produce the final weight. For example, a user with Trust=0.81, Anomaly=0.05, and Experience=0.72 would have a weight of $0.81 \times 0.95 \times 0.72 = 0.554$. This transparency is a deliberate design choice — we believe that users should understand why their votes carry a particular weight, and seeing the decomposition encourages behaviors that build trust (consistent voting, stable location, active participation).

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4.4.5 Simulation Dashboard



Fig 4.6: Simulation dashboard showing phase selector, metric cards (Accuracy, Attack Success, Brier Score, Weight Correlation), color-coded user table, D3.js force-directed network graph, and post credibility list

The simulation dashboard (Fig 4.6) provides a comprehensive view of system performance during simulation runs. The top row contains phase selector buttons and summary metric cards. The middle section displays a color-coded user table (green for honest, red for bots, yellow for adversarial) with columns for all computed scores, alongside an interactive D3.js force-directed network graph that visualizes the user interaction graph. Coordinated bot clusters are visually apparent as tightly connected red node groups. The bottom section shows a post credibility list with horizontal bars indicating each post's final credibility score. This dashboard was invaluable during development — it allowed us to visually inspect the system's behavior and quickly identify issues like insufficient coordination detection or overly aggressive anomaly scoring.

4.5 FUNCTIONAL TESTING

Beyond simulation-based evaluation, we conducted comprehensive functional testing of all system components. The following table summarizes the test scenarios and results:

Table 4.3: Functional Test Results

Test Scenario	Expected Outcome	Result
User Registration	Account created with linked user profile, default Bayesian priors initialized	Passed
User Login	Valid JWT token returned, session established	Passed
Protected Post Creation	Post created under authenticated user with correct author attribution	Passed
Duplicate Voting Prevention	Second vote on same post by same user rejected with 400 error	Passed
Feed Retrieval	All persisted posts returned with computed credibility scores	Passed
Profile State Display	Trust signals, weight decomposition, and activity history shown correctly	Passed
Database Startup Validation	Application fails gracefully when required tables are missing	Passed
React Frontend Build	Production bundle builds successfully with no errors	Passed
Map Rendering	Leaflet map loads with correct markers, radius circles, and popups	Passed
Analytics Endpoint	Returns credibility distribution, user statistics, and system health metrics	Passed
Cross-Origin Requests	CORS configured correctly, frontend communicates with backend from different origin	Passed
Token Expiration	Expired JWT tokens correctly rejected, user prompted to re-authenticate	Passed

All 12 functional tests passed successfully, confirming that the system operates correctly under normal usage conditions.

4.6 STRESS TEST RESULTS

To evaluate system robustness under extreme conditions, we conducted three stress test scenarios that push the system well beyond normal operating parameters:

Table 4.4: Stress Test Scenarios and Results

Scenario	Configuration	Accuracy	Attack Success
Adversarial Majority	64% adversarial users (45 out of 70)	1.000	0.000
Bot Swarm	50% bots (35 out of 70) in 7 coordinated groups	1.000	0.000
Low Activity	Only 60 interactions (avg ~1.2 per post)	1.000	0.000

These results are quite remarkable. Even when adversarial users constitute a clear majority (64%), the system maintains perfect accuracy. This is because the Bayesian confidence correction ensures that the large number of adversarial accounts — which are typically new or have inconsistent track records — carry very low weights. Their numerical advantage is neutralized by their individually low influence.

The bot swarm scenario is similarly robust. Even with 35 bots organized in 7 coordinated groups (each group of 5 bots acting in concert), the coordination detection module identifies and dampens the bot clusters, and the trust propagation ensures that distrust flows through the entire bot network.

The low activity scenario tests the system under data-sparse conditions. With only 60 total interactions across 50 posts, each post has barely more than one vote on average. Despite this extreme sparsity, the system maintains perfect accuracy, largely due to the conservative priors (starting at 0.5) which prevent the system from making confident wrong assessments when evidence is scarce.

4.7 DISCUSSION

The experimental results demonstrate several important properties of the NCPS architecture:

Progressive Improvement: Each phase contributed measurable improvements to at least one metric, validating the modular, phased development approach. The most significant improvements came from ML augmentation (Phase 5), which closed the remaining accuracy gap, and graph trust (Phase 3), which introduced coordination detection capabilities.

Multi-Signal Robustness: The system's resilience against diverse attack types arises from the multiplicative combination of independent signals. An attacker must simultaneously evade Bayesian reliability tracking, graph coordination detection, spatial anomaly detection, behavioral analysis, AND ML-based pattern recognition. Compromising any single dimension is insufficient because the remaining dimensions still expose the attack.

Conservative Design: The safety constraint ($T_i \leq R^*_i$) and the 5-condition propagation gate ensure that the system errs on the side of caution. In practice, this means some true posts may experience slightly delayed propagation, but no false posts gain wide distribution — a trade-off that we believe is appropriate for a system designed to protect communities from misinformation.

Limitations: The system is validated through simulation only. Real-world user behavior may differ from simulated patterns in ways that affect performance. Additionally, the current implementation does not analyze content (text, images) for signs of fabrication, relying entirely on network-level signals. Integration of content analysis could further improve performance, particularly for cases where the social signal is ambiguous.

CHAPTER 5

CONCLUSION AND FUTURE WORKS

5.1 SUMMARY OF CONTRIBUTIONS

This project has presented the Network-aware Credibility and Propagation System (NCPS), a comprehensive platform that replaces engagement-driven content distribution with credibility-driven assessment for local news verification. The key contributions of this work are:

1. A Credibility-Driven Content Distribution Architecture Unlike existing social media systems that amplify content based on engagement metrics, the NCPS fundamentally redefines how content visibility is determined. Posts gain wider reach only when they meet stringent credibility, evidence, consistency, and geographic trust thresholds. This represents a paradigm shift from "what is popular" to "what is likely true" as the basis for information distribution.

2. Multi-Dimensional User Trust Assessment: We developed a user weight model that integrates five independent dimensions — Bayesian reliability with confidence correction, logarithmic experience weighting, five-signal anomaly detection, graph-based trust propagation with coordination detection, and spatial trust with location confidence. The multiplicative composition of these dimensions ensures that compromise along any single dimension is sufficient to suppress a malicious user's influence.

3. Conservative Propagation Control: The 5-condition AND gate for propagation decisions embodies a conservative design philosophy where the cost of spreading false information is treated as strictly higher than the cost of delaying true information. This asymmetric treatment is, to our knowledge, novel in the context of community-based news verification systems.

4. Progressive Architecture Validation: The six-phase development approach allowed us to empirically measure the marginal contribution of each system component. This is valuable not only for understanding NCPS specifically, but also as a methodological contribution — it demonstrates how complex trust-based systems can be developed and validated incrementally.

5. Full-Stack Implementation: Beyond the theoretical contributions, the NCPS is implemented as a complete, functional web application with authentication, rich user interfaces, interactive maps, simulation capabilities, and a comprehensive REST API. The system is not merely a simulation — it is a deployable platform.

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5.2 KEY FINDINGS

The experimental evaluation yielded several notable findings:

Finding 1: Bayesian Estimation Provides a Strong Baseline. Even the simplest configuration (Phase 1 with only Bayesian credibility and confidence correction) achieved 0.900 accuracy. This confirms that the "users as noisy sensors" model, combined with proper uncertainty handling through confidence correction, is a sound foundation for credibility estimation. However, the 0.150 attack success rate in Phase 1 highlights the insufficiency of pure statistical methods when facing coordinated adversaries.

Finding 2: Graph Trust is Essential for Coordination Resistance. The introduction of graph-based trust propagation and coordination detection in Phase 3 was the most architecturally significant addition. It provided the foundation for detecting organized manipulation by identifying clusters of users with suspiciously synchronized behavior. Without graph trust, the system treats each user independently and cannot leverage the structural information embedded in interaction patterns.

Finding 3: Multiple Independent Signals are More Robust Than Any Single Signal. The progressive improvement in anomaly recall across phases (0.000 → 0.480 → 0.680 → 0.800 → 0.840) demonstrates that each new signal category (graph, spatial, behavioral) catches different subsets of malicious users. No single signal is sufficient, but their combination provides comprehensive coverage.

Finding 4: ML Augmentation Closes the Accuracy Gap. The jump from 0.920 to 1.000 accuracy in Phase 5 demonstrates that ML models can learn patterns in the feature space that pure Bayesian estimation misses. The ML component acts as a corrective layer that catches edge cases where statistical aggregation alone produces ambiguous results.

Finding 5: The System is Remarkably Robust Under Stress. The stress test results — maintaining perfect accuracy even with 64% adversarial users, 50% bots, or only 60 interactions — demonstrate that the system's conservative design pays dividends under extreme conditions. The confidence correction mechanism is particularly crucial in these scenarios, preventing masses of low-evidence malicious accounts from overwhelming the system.

Finding 6: Trust Can Only Decrease, Never Inflate. The safety constraint $T_i \leq R^*_i$ is a simple rule but has profound implications. It means that no amount of social engineering (building connections with trusted users) can boost a malicious user's trust above what their individual behavior warrants. This one-directional trust flow is key to the system's attack resistance.

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5.3 FUTURE WORK

While the current NCPS implementation demonstrates strong performance in simulation, several directions for future work could significantly enhance its practical utility:

1. Real-World Dataset Validation: The current evaluation uses synthetic simulation data. Validating the system with real-world datasets from platforms like Twitter, Reddit, or WhatsApp would provide much stronger evidence of practical effectiveness. Particular challenges include obtaining ground truth labels for local news and modeling realistic user population distributions.

2. Content Analysis Integration: The current system does not analyze the textual or visual content of posts. Integrating NLP-based credibility signals (sentiment analysis, claim verification, source extraction) and image forensics (manipulated image detection) would add a powerful complementary dimension to the network-based signals.

3. Administrative Moderation Interface: A dedicated admin panel for human moderators to review flagged content, adjust credibility scores, and manage user accounts would be essential for real-world deployment. The admin interface should provide tools for escalation, appeal handling, and policy enforcement.

4. Push Notifications and Real-Time Alerts: Currently, users must actively visit the application to see updates. Implementing push notifications for high-urgency, high-credibility posts in the user's vicinity would significantly increase the system's utility for time-sensitive local events like emergencies or public safety incidents.

5. Advanced Graph Neural Networks: Replacing the current EigenTrust-inspired propagation with more sophisticated GNN architectures (GraphSAGE, GAT) could improve trust estimation, particularly in sparse graphs where limited edge information makes simple propagation less effective.

6. Privacy-Preserving Location: The current location system collects and stores user coordinates, which raises privacy concerns. Implementing differential privacy mechanisms or zero-knowledge proofs for proximity verification would allow spatial trust computation without exposing exact user locations.

7. Explainable AI Module: Adding explanations for credibility assessments (e.g., "This post has low credibility because 72% of its supporters exhibit anomalous behavior") would increase user trust in the system and help educate users about misinformation patterns.

8. Mobile Application: Developing native mobile apps (iOS and Android) would dramatically increase the system's reach and usability for local news reporting, where users are typically on mobile devices at event locations.

9. Multi-Language Support: Extending the system to support multiple languages, particularly Indian languages (Hindi, Urdu, Kashmiri), would be crucial for deployment in the NIT Srinagar region and across India more broadly.

10. Continuous Threshold Learning: Currently, the system uses fixed thresholds (e.g., anomaly threshold = 0.3, propagation credibility threshold = 0.6). Implementing adaptive threshold learning that adjusts these values based on observed performance could improve the system's adaptability to different community dynamics.

11. Observability and Monitoring: For production deployment, comprehensive monitoring, logging, and alerting infrastructure would be needed. This includes dashboards for system administrators to track credibility distributions, detect systematic attacks, and monitor system health in real-time.

12. Federation: Enabling multiple NCPS instances across different geographic regions to share trust information (while maintaining local autonomy) could create a federated verification network that combines the benefits of local community knowledge with broader credibility signals.

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APPENDIX A

COMPLETE API ENDPOINT REFERENCE

The following table lists all REST API endpoints exposed by the NCPS backend, organized by functional category:

S.No	Route	Method	Auth Required	Description
1	/api/auth/register	POST	No	Register a new user account with email and password
2	/api/auth/login	POST	No	Authenticate and receive JWT token
3	/api/auth/me	GET	Yes	Get current authenticated user details
4	/api/users/	GET	No	List all users with basic profile information
5	/api/users/{user_id}	GET	No	Get detailed user profile including trust signals
6	/api/users/{user_id}/state	GET	No	Get user's current computed state (R, R*, Conf, Exp, etc.)
7	/api/posts/	POST	Yes	Create a new news report with content and location
8	/api/posts/	GET	No	List all posts with computed credibility scores
9	/api/posts/{post_id}	GET	No	Get detailed post information including voter breakdown
10	/api/posts/{post_id}/vote	POST	Yes	Cast a vote (+1 or -1) on a post
11	/api/feed	GET	No	Get credibility-ranked feed of posts
12	/api/map/data	GET	No	Get geographic data for map visualization (posts, users, radii)
13	/api/profile/{user_id}	GET	Yes	Get authenticated user's full profile with trust decomposition
14	/api/analytics/overview	GET	No	Get system-wide analytics (credibility distribution, user stats)
15	/api/simulation/run	POST	No	Trigger a simulation run with specified parameters
16	/api/simulation/results	GET	No	Get results of the last simulation run
17	/api/health	GET	No	Health check endpoint returning system status

Authentication: Endpoints marked with "Auth Required = Yes" expect a valid JWT token in the Authorization header with the format: `Authorization: Bearer <token>`. Tokens are obtained through the `/api/auth/login` endpoint and have a configurable expiration period (default: 24 hours).

Error Handling: All endpoints return standard HTTP status codes: - 200: Success - 201: Created (for POST operations) - 400: Bad Request (invalid input, duplicate vote, etc.) - 401: Unauthorized (missing or invalid token) - 404: Not Found (invalid user_id or post_id) - 500: Internal Server Error

APPENDIX B

HYPERPARAMETER REFERENCE

The following table lists all configurable hyperparameters in the NCPS system, their default values, and descriptions:

Table B.1: Complete Hyperparameter Reference

Parameter Name	Default Value	Description
User Engine		
α_0 (prior_alpha)	1.0	Initial positive Bayesian evidence count
β_0 (prior_beta)	1.0	Initial negative Bayesian evidence count
k (evidence_growth_rate)	0.1	Rate parameter for confidence saturation
E_max (max_experience_count)	100	Maximum experience normalization value
Post Engine		
α_0 (post_prior_alpha)	1.0	Post credibility prior (positive)
β_0 (post_prior_beta)	1.0	Post credibility prior (negative)
λ_{decay} (temporal_decay_rate)	0.01	Exponential decay rate for vote aging
Graph Engine		
λ (damping_factor)	0.5	Trust propagation damping factor
max_iterations	50	Maximum trust propagation iterations
tolerance (ϵ)	1e-6	Convergence tolerance for trust propagation
coord_threshold	0.7	Coordination detection threshold
link_dampen_factor	0.1	Edge weight reduction for coordinated links
Spatial Engine		
σ (proximity_sigma_km)	5.0	Gaussian proximity standard deviation (km)
max_devices	5	Maximum normal device count
max_ips	10	Maximum normal IP address count
base_radius_km	2.0	Initial post propagation radius
max_radius_km	50.0	Maximum allowed propagation radius
radius_step_km	0.5	Radius expansion step per evaluation
ML Engine		
α_{ml} (ml_weight)	0.15	ML prediction weight in final credibility
γ_{mem} (memory_weight)	0.10	Memory score weight in final credibility
min_training_samples	10	Minimum resolved posts before ML training
Decision Engine		
credibility_threshold	0.6	Minimum credibility for propagation
evidence_threshold	3.0	Minimum evidence mass for propagation
variance_threshold	0.25	Maximum variance for propagation
min_age_seconds	60	Minimum post age for propagation
location_trust_threshold	0.3	Minimum mean voter location trust
Anomaly Detection		

Parameter Name	Default Value	Description
anomaly_threshold	0.3	Score above which user is considered anomalous
burst_window_seconds	60	Time window for burst detection
max_actions_per_window	10	Maximum normal actions per burst window
Simulation		
num_honest	40	Default honest user count
num_noisy	5	Default noisy user count
num_adversarial	5	Default adversarial user count
num_bots	20	Default bot count
num_posts	50	Default post count
num_interactions	1000	Default interaction count

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APPENDIX C

SETUP AND EXECUTION GUIDE

This appendix provides instructions for setting up and running the NCPS platform on a local development environment.

C.1 PREREQUISITES

Ensure the following software is installed: - **Python** 3.10 or higher - **Node.js** 18+ and **npm** 9+ - **PostgreSQL** 14+ - **Git** for version control

C.2 BACKEND SETUP

```
# 1. Clone the repository
git clone https://github.com/Niteshkrjhag/NCPS.git
cd NCPS

# 2. Create a Python virtual environment
python3 -m venv venv
source venv/bin/activate      # Linux/macOS
# or: venv\Scripts\activate   # Windows

# 3. Install backend dependencies
pip install -r requirements.txt

# 4. Configure environment variables
cp .env.example .env
# Edit .env with your PostgreSQL credentials:
# DATABASE_URL=postgresql://user:password@localhost:5432/ncps_db
# SECRET_KEY=your-secret-key-here
# ALGORITHM=HS256
# ACCESS_TOKEN_EXPIRE_MINUTES=1440
```

C.3 DATABASE SETUP

```
# 1. Create the PostgreSQL database
createdb ncps_db

# 2. Initialize database schema
# The application auto-creates tables on first run via SQLAlchemy
python -c "from backend.app.database import engine, Base;
Base.metadata.create_all(bind=engine)"
```

C.4 BACKEND EXECUTION

```
# Start the FastAPI server
cd backend
uvicorn app.main:app --reload --host 0.0.0.0 --port 8000

# The API is now accessible at http://localhost:8000
# API documentation at http://localhost:8000/docs (Swagger UI)
# Alternative docs at http://localhost:8000/redoc
```

C.5 FRONTEND SETUP (VANILLA JAVASCRIPT)

```
# The vanilla JS frontend is served as static files
# Open directly in browser or use a simple HTTP server
cd frontend
python3 -m http.server 5500

# Access at http://localhost:5500
```

C.6 REACT FRONTEND SETUP

```
# 1. Install React dependencies
cd react-frontend
npm install

# 2. Start development server
npm run dev

# Access at http://localhost:5173
```

C.7 RUNNING THE SIMULATION

```
# Option 1: Run via API endpoint
curl -X POST http://localhost:8000/api/simulation/run

# Option 2: Run directly via Python
python -m backend.app.simulation.run_simulation

# Option 3: Use the web-based simulation dashboard
# Navigate to /simulation in the webapp
```

C.8 RUNNING TESTS

```
# Run the full test suite
cd backend
python -m pytest tests/ -v

# Run specific test categories
python -m pytest tests/test_user_engine.py -v
python -m pytest tests/test_post_engine.py -v
python -m pytest tests/test_graph_engine.py -v
python -m pytest tests/test_simulation.py -v
```

C.9 PROJECT DIRECTORY STRUCTURE

```

NCPS/
├── backend/
│   ├── app/
│   │   ├── main.py           # FastAPI application entry point
│   │   ├── database.py       # Database connection and session management
│   │   ├── models.py         # SQLAlchemy ORM models
│   │   ├── auth.py           # JWT authentication logic
│   │   └── engine/
│   │       ├── user_engine.py # User weight computation
│   │       ├── post_engine.py # Post credibility computation
│   │       ├── graph_engine.py # Graph trust and coordination detection
│   │       ├── spatial.py     # Spatial trust and propagation
│   │       ├── ml_engine.py   # ML and memory augmentation
│   │       ├── signal_engine.py # Behavioral signal computation
│   │       ├── urgency.py     # Urgency classification
│   │       └── decision.py    # Propagation decision engine
│   ├── routes/
│   │   ├── auth.py           # Authentication endpoints
│   │   ├── users.py          # User management endpoints
│   │   ├── posts.py          # Post and voting endpoints
│   │   ├── feed.py           # Feed retrieval endpoints
│   │   ├── map.py            # Map data endpoints
│   │   └── analytics.py      # Analytics endpoints
│   ├── simulation/
│   │   └── run_simulation.py  # Simulation framework
│   └── webapp/
│       ├── static/           # Static files (CSS, JS, images)
│       └── templates/        # HTML templates
├── tests/
│   └── requirements.txt      # Test suite
├── requirements.txt          # Python dependencies
├── frontend/
│   └── index.html            # Vanilla JS frontend
├── css/
├── js/
├── react-frontend/          # React frontend
│   ├── src/
│   ├── package.json
│   └── vite.config.js
├── docs/                    # Documentation
│   └── context/              # Project context files
├── CHANGELOG.md
├── hyperparameter.md
└── README.md
    
```